
MATHEMATICAL BIOLOGY

How Does a Hydra Decide Where to Grow a Head?

Chemical and mechanical routes to pattern formation

From regeneration to Turing instability, springs and energy



LECTURE MAP

The road from Hydra to pattern formation

From biological observations to mathematical mechanisms.

01 The biological puzzle

02 Reaction–diffusion approach

03 Turing instability

04 Gierer–Meinhardt model

05 Growth, wounds & long-range dynamics

06 Mechanochemical pattern formation



01 THE BIOLOGICAL PUZZLE

Hydra as a model organism

A simple organism makes symmetry breaking directly visible.



Hydra: simple body, continuous renewal

- **A small freshwater cnidarian**

A transparent, tube-shaped polyp attached to the substrate by a basal disc.

- **Simple oral–aboral anatomy**

A head with hypostome and tentacles, a body column, a budding zone and a foot.

- **Two epithelial layers**

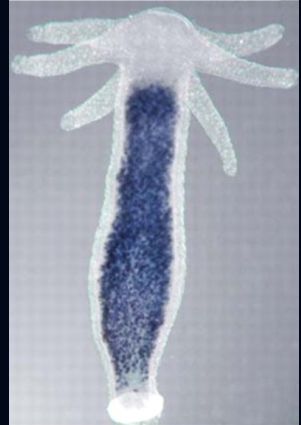
Epidermis outside and gastrodermis inside, separated by mesoglea around one gastric cavity.

- **Continuous tissue renewal**

Stem cells divide in the body column while cells are displaced toward the head and foot.

- **Evolutionary significance**

An early-branching animal model for organizers, axial patterning and regeneration.



Augustin R et al., Dev Biol. 2006; 296(1):62-70.

Hydra can rebuild a whole animal from very little

- **Head removal**

The remaining body column regenerates a new head at the wound.

- **Bisection**

The apical half rebuilds a foot; the basal half rebuilds a head — producing two complete animals.

- **Small tissue fragment**

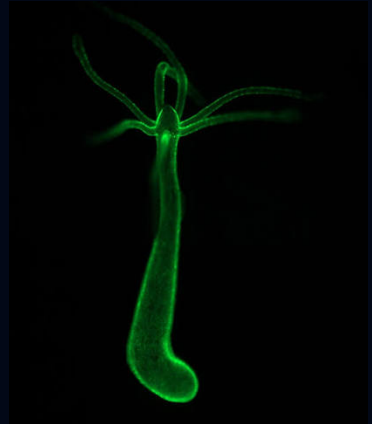
A fragment closes, rounds up, breaks symmetry and reconstructs an oral–aboral axis.

- **Dissociated cells**

Cells can be separated, reaggregated, sorted into layers and self-organized into new polyps.

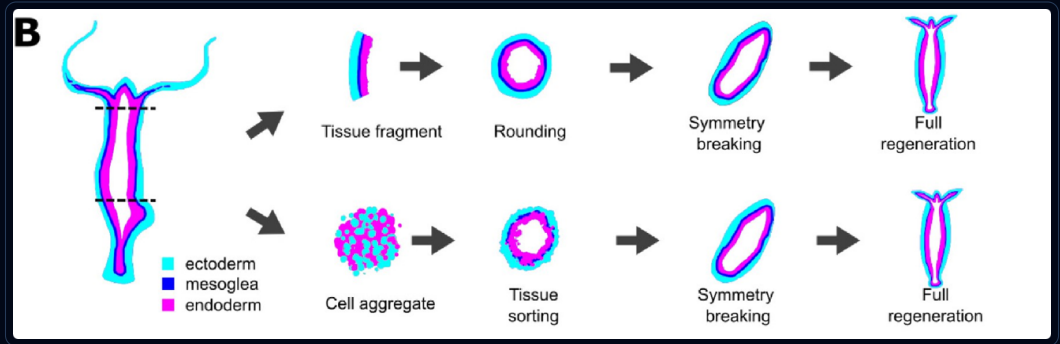
- **The central puzzle**

Where does the new head form when the starting tissue is nearly symmetric?



Wittlieb J et al., Dev Biol. 2006; 296(1):62-70.

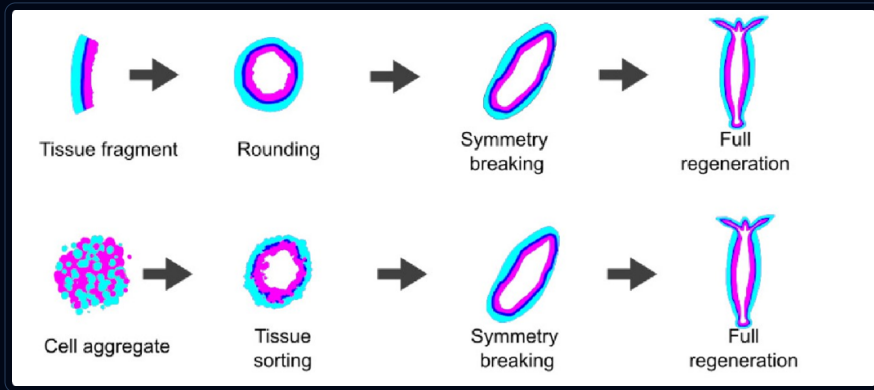
Regeneration unfolds through ordered stages



Wang, Bialas, Goel & Collins, *Integrative and Comparative Biology* 63 (2023), Fig. 1B.

Very different initial conditions pass through a similar spheroid stage before rebuilding an body axis.

Symmetry breaks before a new head becomes visible



Wang, Bialas, Goel & Collins, *Integrative and Comparative Biology* 63 (2023), Fig. 1B.

A local site acquires head-forming identity first, visible tentacles and the complete head-foot axis appear later

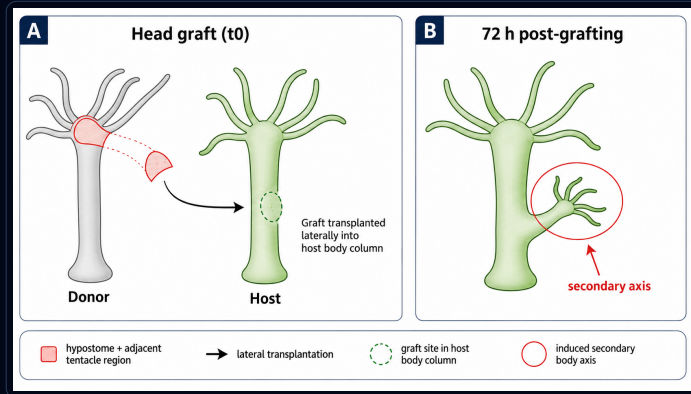
01 THE BIOLOGICAL PUZZLE

Hydra grafting experiments

Transplantation reveals that organizer activity depends on position and context.



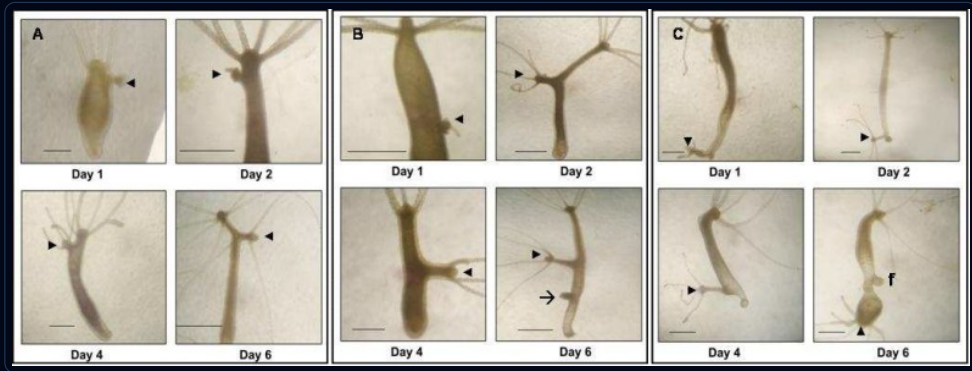
A head graft can organize a second body axis



Schematic based on the classic hypostome transplantation experiment of Browne (1909); see also Broun & Bode (2002).

The graft does not simply grow into a second Hydra, it recruits a new body axis.

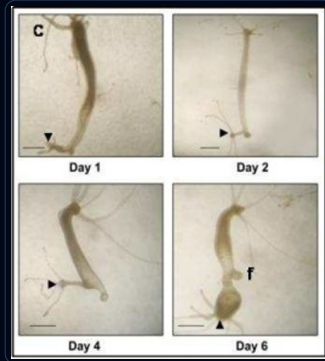
Impact of the grafting position



Kadu, Ghaskadbi & Ghaskadbi, *International Journal of Molecular and Cellular Medicine* 1 (2012).

The same graft can produce different outcomes depending on its position

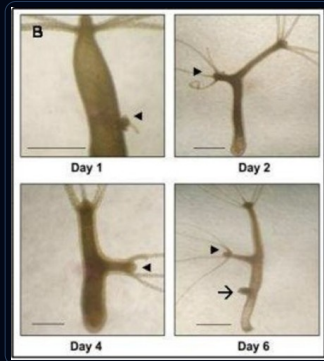
Far from the host head: a complete second axis



Kadu, Ghaskadbi & Ghaskadbi, *Int. J. Mol. Cell. Med.* 1 (2012), Fig. 2C.

- **A complete secondary axis**
The induced structure develops a foot, body column, hypostome and tentacles.
- **The strongest response**
Far from the host head, the same graft produces the longest and most complete secondary axis.

Mid-body graft: an intermediate second axis



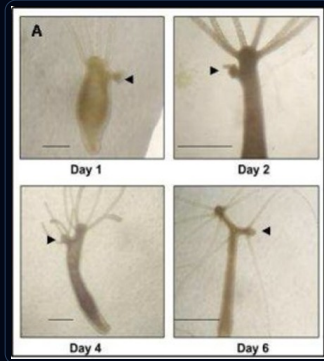
Kadu, Ghaskadbi & Ghaskadbi, *Int. J. Mol. Cell. Med.* 1 (2012), Fig. 2B.

- **An intermediate secondary axis**
The induced axis grows substantially longer than after an upper-body graft.
- **Only position has changed**
The donor tissue is the same; moving the graft along the host axis changes the resulting morphology.

Near the host head: only a short second axis



upper host position



Kadu, Ghaskadbi & Ghaskadbi, *Int. J. Mol. Cell. Med.* 1 (2012), Fig. 2A.

- **A very short secondary axis**

The hypostome graft still induces a new head, but only a short body column develops.

- **Growth is strongly constrained**

Close to the host head, the induced axis remains small despite using the same organizer tissue.

Multiple grafts can impose multiple axes

- **Three grafts on one host**

A complete foot was grafted in the upper region, hypostome pieces were grafted in the middle and lower regions.

- **Three secondary axes**

All three grafted regions produced an induction simultaneously.

- **Organizer identity matters**

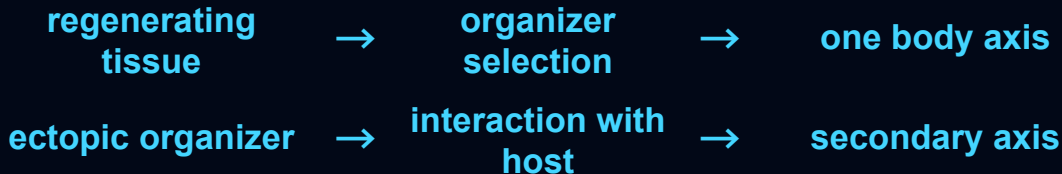
The axes induced by hypostome tissue were substantially longer than the axis induced by the foot graft.



Kadu, Ghaskadbi & Ghaskadbi, *Int. J. Mol. Cell. Med.* 1 (2012), Fig. 6.

Multiple local organizers can coexist and impose several body axes on the same tissue.

From experiments to a mathematical model



- **Spontaneous pattern formation**

How can a localized organizer emerge from weakly patterned or nearly symmetric tissue?

- **Competition and uniqueness**

Why does normal regeneration usually select one head, while several grafts can sustain several axes?

- **Organizer induction**

Why can a transplanted organizer recruit host tissue and initiate a new axis?

- **Context dependence**

Why does the same hypostome graft produce different secondary axes at different positions in the host?

LECTURE MAP

The road from Hydra to pattern formation

From biological observations to mathematical mechanisms.

01 The biological puzzle

02 **Reaction–diffusion approach**

03 Turing instability

04 Gierer–Meinhardt model

05 Growth, wounds & long-range dynamics

06 Mechanochemical pattern formation



02 REACTION-DIFFUSION APPROACH

Chemical patterning

Chemical signals provide a natural
mechanism for spatial organization.



Could chemical signals determine where a head forms?

- **Cells communicate through chemical signals**

Signaling molecules can be produced, transformed and removed by cells.

- **Concentrations can vary in space**

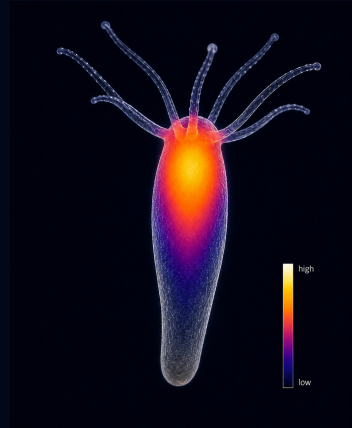
Different regions of the tissue may therefore experience different levels of the same signal.

- **A local maximum can select a site**

A sufficiently high concentration may trigger head-forming activity in one particular region.

- **Chemistry can position the organizer**

The location of head formation becomes a question of how a spatial concentration pattern arises.



Reactions change signals, diffusion moves them

REACTION

- **Local chemical change**

Cells can produce, remove or transform signaling molecules.

- **Local feedback**

The production of a signal may depend on the concentrations already present in the same region.

DIFFUSION

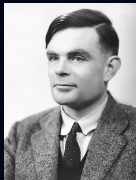
- **Spatial redistribution**

Signaling molecules spread from regions of higher concentration toward regions of lower concentration.

- **Communication across tissue**

Diffusion couples nearby regions, allowing a local chemical change to influence its surroundings.

Mechanism of chemical pattern formation



1952
Alan Turing



- **A homogeneous tissue need not stay homogeneous**
Chemical substances can react and diffuse through tissue, allowing spatial structure to emerge from small perturbations.
- **Diffusion becomes part of the mechanism**
Spatial transport is not merely smoothing: together with reactions it can participate in pattern formation.
- **Hydra was already on the list**
Turing explicitly mentioned stationary patterns as a possible explanation for tentacle organization in Hydra.

Reaction and diffusion can, in principle, turn an almost homogeneous state into spatial structure.

Gierer and Meinhardt propose a model



Alfred Gierer



Hans Meinhardt

A. GIERER & H. MEINHARDT / 1972

A Theory of Biological Pattern Formation

A mechanism for generating spatial organization in an initially almost homogeneous tissue.

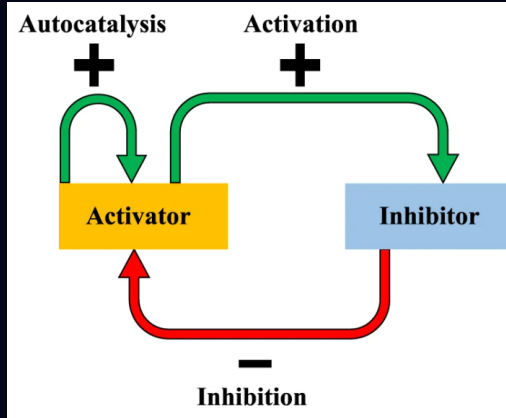
- **Local self-activation**

A small local excess reinforces itself and can develop into a localized organizing region.

- **Long-range inhibition**

The activated region generates an inhibitory effect that acts over a larger spatial range.

The activator–inhibitor idea



- **Activator**

Enhances its own production and promotes local organizer formation. Its effect remains relatively localized.

- **Inhibitor**

Is produced by activated regions, suppresses further activation and acts over a longer spatial range.

The Gierer–Meinhardt model

$$\partial_t u = \underbrace{D_u \Delta u}_{\text{diffusion}} + \underbrace{\frac{au^2}{K + v} - \mu_u u}_{\text{reaction}}$$

$$\partial_t v = \underbrace{D_v \Delta v}_{\text{diffusion}} + \underbrace{bu^2 - \mu_v v}_{\text{reaction}}$$

- **u : activator**
Promotes its own production and initiates local organizer activity

- **v : inhibitor**
Produced by the activator and suppresses its production

D_u, D_v diffusion rates, typically $D_v > D_u$

a strength of activator self-amplification

b strength of inhibitor production

K scale of inhibition by v

μ_u activator degradation rate

μ_v inhibitor degradation rate

A general reaction–diffusion system

$$\partial_t u = \underbrace{D_u \Delta u}_{\text{diffusion}} + \underbrace{f(u, v)}_{\text{reaction}},$$

$$\partial_t v = \underbrace{D_v \Delta v}_{\text{diffusion}} + \underbrace{g(u, v)}_{\text{reaction}}.$$

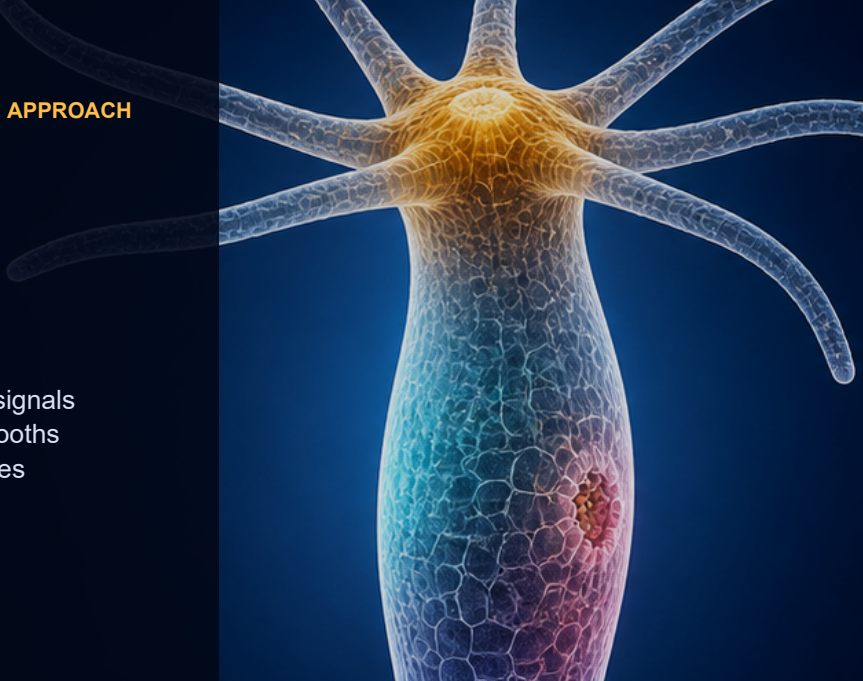
- **u and v**
Concentrations of two interacting chemical signals.
- **f and g**
Possibly nonlinear local production, removal and interaction terms.

We begin by deriving the spatial transport term.

02 REACTION-DIFFUSION APPROACH

Diffusion

Diffusion redistributes signals through space and smooths concentration differences



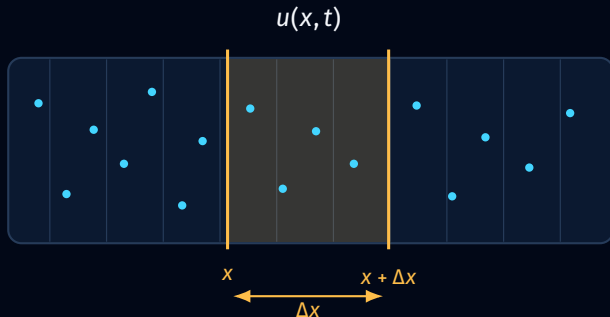
Diffusion begins with a local balance

- Concentration**

$u(x, t)$ measures the amount of a substance per unit length at point x .

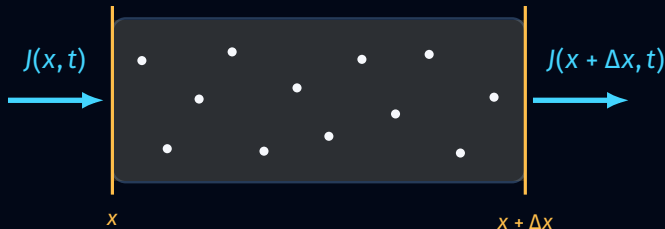
- Control volume**

We track the substance inside a short interval of length Δx .



$$\text{amount in the interval} \approx u(x, t) \Delta x$$

Boundary flux



Flux J : amount of substance crossing a boundary.

$J(x, t) > J(x + \Delta x, t)$
amount increases

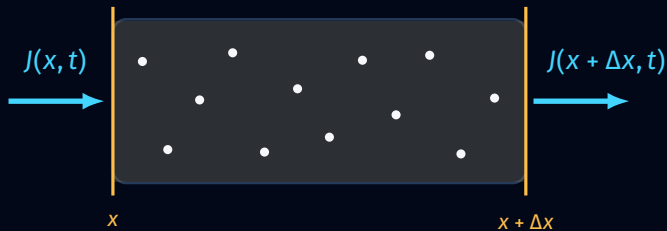
$J(x, t) = J(x + \Delta x, t)$
amount stays constant

$J(x, t) < J(x + \Delta x, t)$
amount decreases

$$\frac{u(x, t + \Delta t) \Delta x - u(x, t) \Delta x}{\Delta t} = J(x, t) - J(x + \Delta x, t)$$

Only a difference between the boundary fluxes changes the amount inside.

The local conservation law



$$\frac{u(x, t + \Delta t) - u(x, t)}{\Delta t} = - \frac{J(x + \Delta x, t) - J(x, t)}{\Delta x}$$

$$\Downarrow \quad \Delta t, \Delta x \rightarrow 0$$

$$\boxed{\partial_t u = -\partial_x J}$$

Fick's law of diffusion



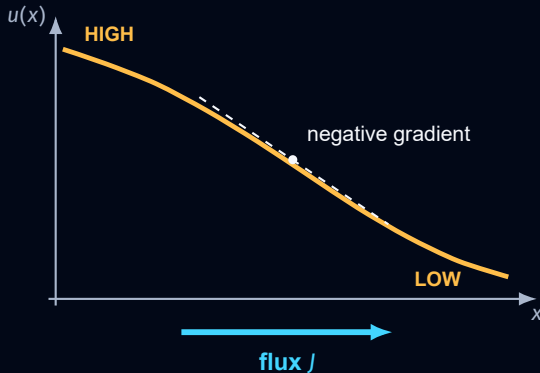
ADOLF FICK
1829–1901

1855

Ueber Diffusion

Using Fourier's heat-conduction analogy, Fick related diffusive flux to the concentration gradient

$$J = -D \partial_x u$$



$$\partial_x u < 0 \Rightarrow J > 0$$

flux to the right

$$\partial_x u = 0 \Rightarrow J = 0$$

no concentration-driven flux

$$|J| = D |\partial_x u|$$

steeper gradient, larger flux

Diffusion equation

LOCAL CONSERVATION

$$\partial_t u = -\partial_x J$$

+

FICK'S LAW

$$J = -D\partial_x u$$



DIFFUSION EQUATION

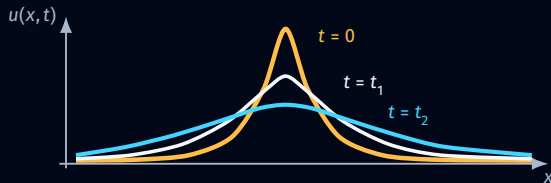
$$\partial_t u = D\partial_{xx} u$$

Diffusion equation

$$\Omega = [0, L] \subset \mathbb{R}$$

$$\partial_t u(x, t) = D \partial_{xx} u(x, t),$$

$$u(x, 0) = u_0(x).$$



- No-flux boundary (Neumann)**

The sealed ends prevent the substance from leaving



$$\partial_x u(0, t) = 0$$

$$\partial_x u(L, t) = 0$$

- Absorbing boundary (Dirichlet)**

Substance that reaches an open end leaves the domain



$$u(0, t) = 0,$$

$$u(L, t) = 0.$$

Fourier turns diffusion into a problem of spatial modes

1822

Théorie analytique de la chaleur

- Fourier's question**

How does an arbitrary temperature profile evolve under the heat equation?

- The key idea**

Decompose a complicated spatial profile into simple sine and cosine waves.

$$u_0(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right).$$

AN INITIAL PROFILE



decompose

$n = 0$



$n = 1$



$n = 2$



...

Instead of solving for every initial profile separately, solve how each elementary spatial wave evolves.

No-flux boundaries - cosine modes

- A mode keeps its spatial shape**

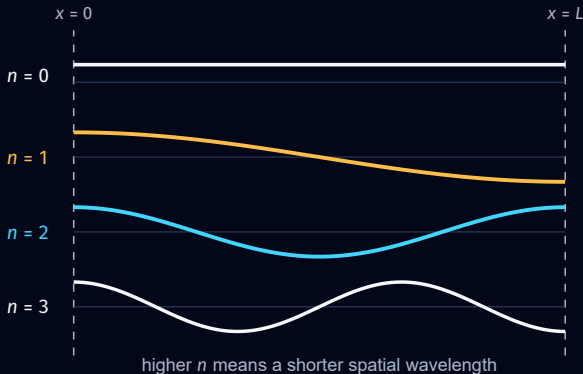
Applying the diffusion operator changes only its amplitude, not the form of the profile.

$$-\phi_n'' = \mu_n \phi_n,$$

$$\phi_n'(0) = \phi_n'(L) = 0.$$

$$\phi_n(x) = \cos\left(\frac{n\pi x}{L}\right),$$

$$\mu_n = \left(\frac{n\pi}{L}\right)^2.$$



The derivative vanishes at both endpoints, so every cosine satisfies the no-flux boundary conditions.

Eigenfunctions and eigenmodes

$$u(x, t) = \sum_{n=0}^{\infty} a_n(t) \phi_n(x)$$

$$\phi_n(x) = \cos\left(\frac{n\pi x}{L}\right)$$

$$\mu_n = \left(\frac{n\pi}{L}\right)^2$$

$$-\phi_n'' = \mu_n \phi_n$$

$$\phi_n'(0) = \phi_n'(L) = 0$$

$$\partial_t u(x, t) = \sum_{n=0}^{\infty} a_n'(t) \phi_n = D \sum_{n=0}^{\infty} a_n(t) \phi_n'' = \partial_{xx} u(x, t)$$

$$\begin{aligned} a_n'(t) &= -D\mu_n a_n(t) \\ a_n(t) &= a_n(0)e^{-D\mu_n t} \end{aligned}$$

$n = 1$: slower decay



$n = 4$: faster decay



$t = 0$

$t = 1$

- Short waves disappear first**

Because $\mu_n = (n\pi/L)^2$, modes with larger n decay much faster.

Diffusion in higher dimensions

$$x \in \Omega \subset \mathbb{R}^d \quad \partial_t u = D \Delta u$$

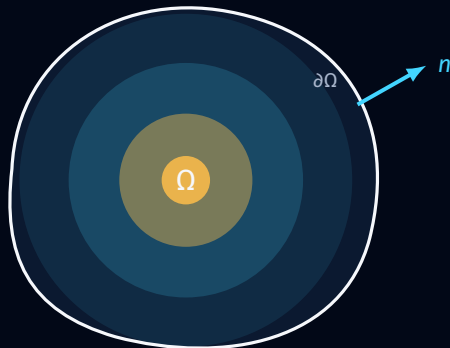
In Cartesian coordinates,

$$\Delta u = \partial_{x_1 x_1} u + \dots + \partial_{x_d x_d} u$$

- No-flux boundary**

A closed tissue does not allow the signal to cross its boundary.

$$\nabla u \cdot n = 0 \quad \text{on } \partial\Omega$$



substance remains inside the tissue

Eigenfunctions on a bounded domain

THEOREM

There exist eigenfunctions $\{\phi_k\}_{k=0}^{\infty}$ and eigenvalues $\{\mu_k\}_{k=0}^{\infty}$ of the Laplace operator satisfying

$$-\Delta \phi_k = \mu_k \phi_k, \quad \nabla \phi_k \cdot n = 0 \quad \text{on } \partial\Omega.$$

$$0 = \mu_0 < \mu_1 \leq \mu_2 \leq \dots, \quad \boxed{\mu_k \rightarrow \infty}.$$

The eigenfunctions form an orthonormal basis, solution can be written as a sum of spatial modes.

$$u_0(x) = \sum_{k=0}^{\infty} a_k(0) \phi_k(x),$$

$$u(x, t) = \sum_{k=0}^{\infty} a_k(0) e^{-D\mu_k t} \phi_k(x).$$

- Different modes, different rates**

Large eigenvalues correspond to rapidly decaying spatial structure.

- No mode is amplified**

Every nonconstant mode decreases under diffusion alone.

02 REACTION-DIFFUSION APPROACH

Local dynamics

Reactions change signals locally
through production, degradation
and nonlinear interactions



Reaction

Begin with a system of ordinary differential equations

$$\partial_t u = f(u, v)$$

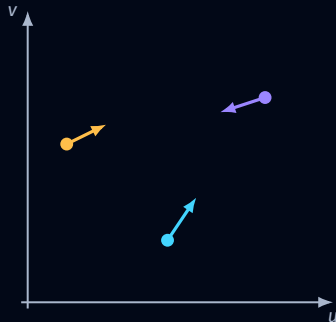
$$\partial_t v = g(u, v)$$

u and v are the local concentrations of two substances

The functions $f(u, v)$ and $g(u, v)$ describe how these concentrations change depending on their current values

- **One state at every position**

The pair (u, v) represents the local chemical state.



the vector $(f(u, v), g(u, v))$ determines the direction of change

Reaction - vector field and trajectory

$$\partial_t u = f(u, v)$$

$$\partial_t v = g(u, v)$$

supplemented with initial conditions

$$u_0 = u(0) \quad v_0 = v(0)$$

- **Initial state**

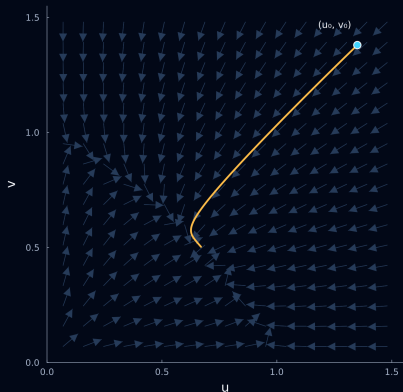
The pair (u_0, v_0) selects the starting point in the concentration plane.

- **Velocity**

The vector $(f(u, v), g(u, v))$ gives the instantaneous direction of change.

- **Trajectory**

The solution curve is tangent to the vector field at every point.



Steady state

At a steady state $u(t) \equiv \bar{u}$, $v(t) \equiv \bar{v}$, both concentrations remain constant in time

$$f(\bar{u}, \bar{v}) = 0$$

$$g(\bar{u}, \bar{v}) = 0$$

NULLCLINES

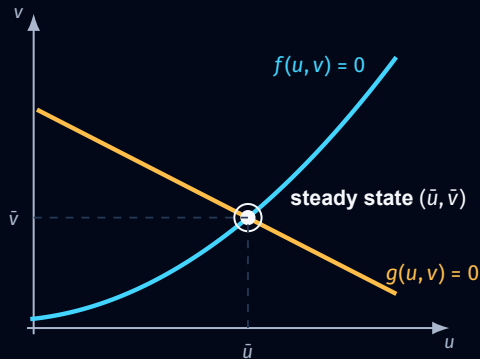
$$f(u, v) = 0$$

u does not change

$$g(u, v) = 0$$

v does not change

Each nullcline contains states for which one concentration is instantaneously stationary



the nullclines organize the local phase plane

Lyapunov Stability

Consider two solutions of the same system

$$\partial_t u = f(u, v)$$

$$\partial_t v = g(u, v)$$

$$u(0) = u_0$$

$$\tilde{u}(0) = \tilde{u}_0$$

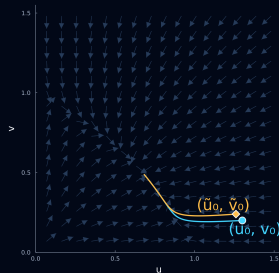
$$v(0) = v_0$$

$$\tilde{v}(0) = \tilde{v}_0$$

LYAPUNOV STABILITY

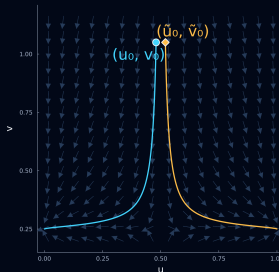
For every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\begin{aligned} \|(u_0, v_0) - (\tilde{u}_0, \tilde{v}_0)\| < \delta &\Rightarrow \\ \|(u(t), v(t)) - (\tilde{u}(t), \tilde{v}(t))\| < \varepsilon &\text{ for all } t \geq 0 \end{aligned}$$



STABLE

Nearby solutions converge



UNSTABLE

Nearby solutions separate

Stability of constant steady state

$$u(0) = \bar{u} + \eta_0$$

$$v(0) = \bar{v} + \xi_0$$

The perturbation is given by

$$\eta(t) = u(t) - \bar{u}$$

$$\xi(t) = v(t) - \bar{v}$$

- Asymptotically stable**

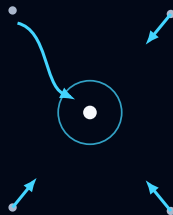
Every sufficiently small perturbation returns to the equilibrium.

$$\begin{aligned} \eta(t) &\longrightarrow 0 \\ \xi(t) &\longrightarrow 0 \end{aligned} \quad \text{as } t \longrightarrow \infty$$

- Unstable**

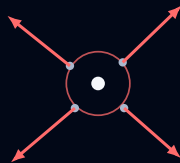
At least one arbitrarily small perturbation moves away.

stable



nearby states return

unstable



some nearby states escape

Linearized stability

Near the steady state $(\bar{u}, \bar{v})$ write

$$u = \bar{u} + \eta, \quad v = \bar{v} + \xi$$

TAYLOR EXPANSION

$$\partial_t \eta = f(\bar{u} + \eta, \bar{v} + \xi) = f(\bar{u}, \bar{v}) + f_u(\bar{u}, \bar{v})\eta + f_v(\bar{u}, \bar{v})\xi + O(\|(\eta, \xi)\|^2)$$

$$\partial_t \xi = g(\bar{u} + \eta, \bar{v} + \xi) = g(\bar{u}, \bar{v}) + g_u(\bar{u}, \bar{v})\eta + g_v(\bar{u}, \bar{v})\xi + O(\|(\eta, \xi)\|^2)$$

JACOBIAN

$$J(\bar{u}, \bar{v}) = \begin{pmatrix} f_u(\bar{u}, \bar{v}) & f_v(\bar{u}, \bar{v}) \\ g_u(\bar{u}, \bar{v}) & g_v(\bar{u}, \bar{v}) \end{pmatrix}$$

NEGLECTING HIGHER-ORDER TERMS

$$\begin{pmatrix} \partial_t \eta \\ \partial_t \xi \end{pmatrix} = J(\bar{u}, \bar{v}) \begin{pmatrix} \eta \\ \xi \end{pmatrix}$$

Close to the steady state, the Jacobian determines the leading-order dynamics.

Eigenvalues of the Jacobian

LINEARIZED SYSTEM

$$\begin{pmatrix} \partial_t \eta \\ \partial_t \xi \end{pmatrix} = J(\bar{u}, \bar{v}) \begin{pmatrix} \eta \\ \xi \end{pmatrix}$$

$$J(\bar{u}, \bar{v}) = \begin{pmatrix} f_u(\bar{u}, \bar{v}) & f_v(\bar{u}, \bar{v}) \\ g_u(\bar{u}, \bar{v}) & g_v(\bar{u}, \bar{v}) \end{pmatrix}$$

The eigenvalues are the roots of

$$\det(J(\bar{u}, \bar{v}) - \lambda I) = 0$$

THEOREM

Let λ_1 and λ_2 be the eigenvalues of $J(\bar{u}, \bar{v})$

- If $\operatorname{Re} \lambda_1 < 0$ and $\operatorname{Re} \lambda_2 < 0$ then the steady state $(\bar{u}, \bar{v})$ is **asymptotically stable**
- If $\operatorname{Re} \lambda_i > 0$ for at least one eigenvalue then the steady state $(\bar{u}, \bar{v})$ is **unstable**

If an eigenvalue has zero real part, linearization alone may be inconclusive

Eigenvalues

$$\begin{pmatrix} \partial_t \eta \\ \partial_t \xi \end{pmatrix} = J(\bar{u}, \bar{v}) \begin{pmatrix} \eta \\ \xi \end{pmatrix}$$

$$J(\bar{u}, \bar{v}) = \begin{pmatrix} f_u(\bar{u}, \bar{v}) & f_v(\bar{u}, \bar{v}) \\ g_u(\bar{u}, \bar{v}) & g_v(\bar{u}, \bar{v}) \end{pmatrix}$$

CHARACTERISTIC EQUATION

$$\det(J - \lambda I) = \begin{vmatrix} f_u - \lambda & f_v \\ g_u & g_v - \lambda \end{vmatrix} = (f_u - \lambda)(g_v - \lambda) - f_v g_u = \lambda^2 - \text{tr}(J)\lambda + \det(J)$$

DETERMINANT

- If $\det(J) < 0$, one eigenvalue is positive and one is negative

The steady state is an unstable saddle

- If $\det(J) > 0$, real eigenvalues have the same sign or form a complex conjugate pair

The trace determines the sign of their real parts

TRACE WHEN $\det(J) > 0$

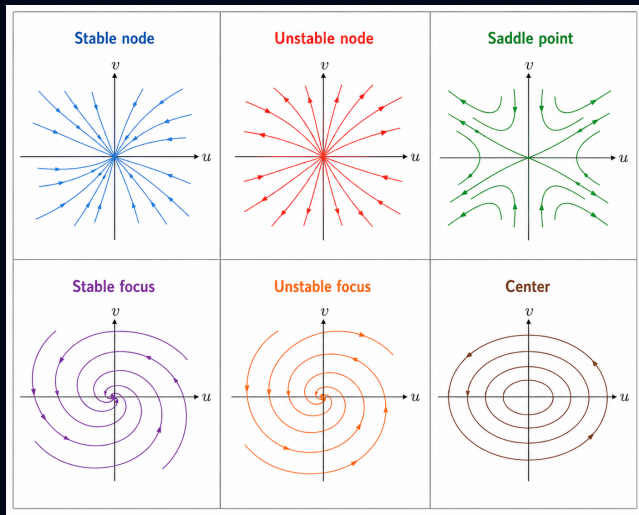
- If $\text{tr}(J) < 0$, both eigenvalues have negative real parts

The steady state is asymptotically stable

- If $\text{tr}(J) > 0$, both eigenvalues have positive real parts

The steady state is unstable

Eigenvalues determine local stability



Trace and determinant classify a two-variable equilibrium

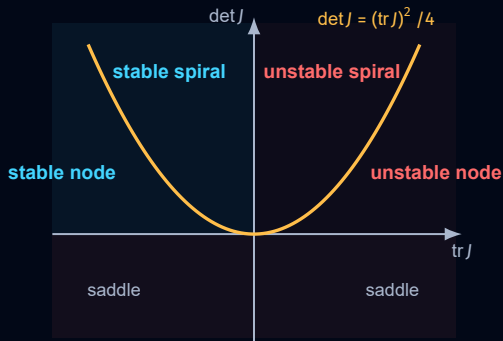
$$\det(J(\bar{u}, \bar{v}) - \lambda I) = 0$$

$$\lambda^2 - \text{tr}(J)\lambda + \det(J) = 0$$

Therefore

$$\lambda_1 + \lambda_2 = \text{tr}(J)$$

$$\lambda_1 \lambda_2 = \det(J)$$



LECTURE MAP

The road from Hydra to pattern formation

From biological observations to mathematical mechanisms.

01 The biological puzzle

02 Reaction–diffusion approach

03 Turing instability

04 Gierer–Meinhardt model

05 Growth, wounds & long-range dynamics

06 Mechanochemical pattern formation



03 TURING INSTABILITY

The Turing idea

Two interacting substances can turn diffusion from a smoothing process into a source of spatial structure



Scalar reaction diffusion equation

$$\partial_t u = D\Delta u + f(u)$$

Let $\bar{u}$ be a stable homogeneous equilibrium

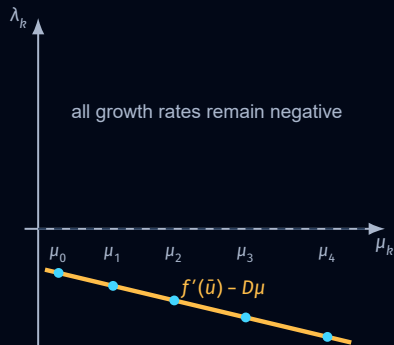
$$f(\bar{u}) = 0 \quad f'(\bar{u}) < 0$$

For a perturbation in the k th spatial eigenfunction

$$u(x, t) = \bar{u} + a_k(t)\phi_k(x)$$

$$\partial_t a_k = (f'(\bar{u}) - D\mu_k) a_k$$

$$\lambda_k = f'(\bar{u}) - D\mu_k < 0$$



- Diffusion adds damping**

Every nonconstant spatial mode is more stable than the homogeneous mode.

Scalar reaction diffusion equation

$$\begin{aligned}\partial_t u &= D\Delta u + f(u) & x \in \Omega \quad t > 0 \\ u(x, 0) &= u_0(x) & x \in \Omega \\ \nabla u \cdot \mathbf{n} &= 0 & x \in \partial\Omega\end{aligned}$$

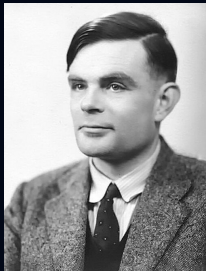
THEOREM

Let $\Omega \subset \mathbb{R}^n$ be a bounded convex domain. Every nonconstant stationary solution of the scalar reaction–diffusion equation with homogeneous Neumann boundary conditions is unstable

Every stable stationary solution is spatially constant

$$u(x) \equiv \bar{u} \quad \text{and} \quad f(\bar{u}) = 0$$

Diffusion driven instability



1952

Alan Turing

The Chemical Basis of Morphogenesis

$$\partial_t u = D_u \Delta u + f(u, v)$$

$$\partial_t v = D_v \Delta v + g(u, v)$$

- **Stable reactions**

Without spatial transport, small perturbations return to the homogeneous equilibrium.

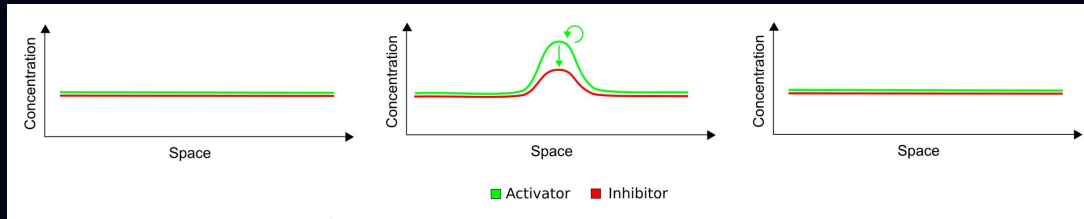
- **Unstable reaction–diffusion system**

After diffusion is introduced, one nonconstant spatial mode can begin to grow.

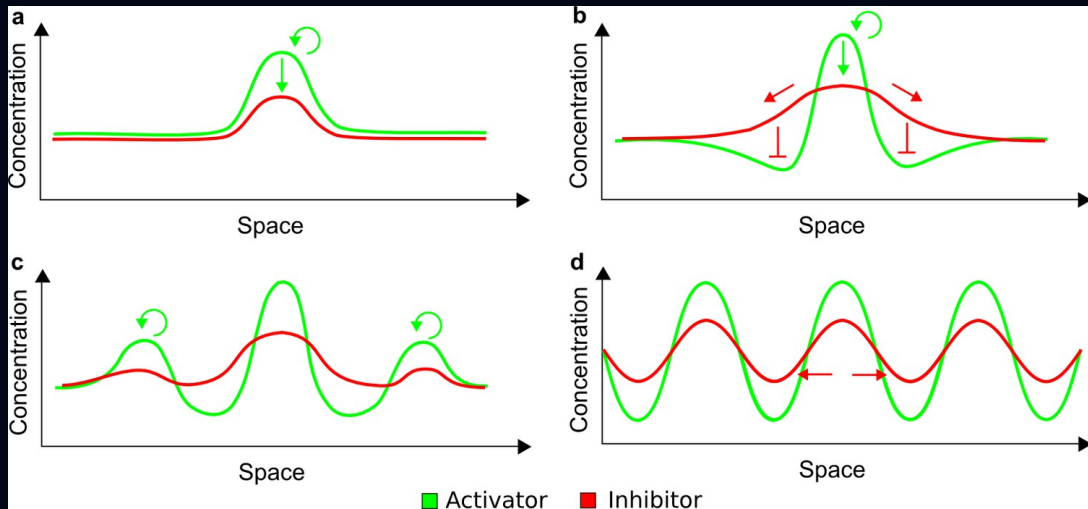
$$D_u \neq D_v$$

Unequal diffusion is essential for classical two-species Turing instability.

Equal diffusion shifts every spatial mode toward stability



Unequal diffusion can amplify a spatial mode



03 TURING INSTABILITY

Stability analysis

Linearization and spatial eigenfunctions
reduce the PDE to one stability
problem for each spatial mode



The full reaction–diffusion problem

Let $\Omega \subset \mathbb{R}^n$ be a bounded domain

$$\begin{aligned} \partial_t u &= D_u \Delta u + f(u, v) & x \in \Omega \quad t > 0 \\ \partial_t v &= D_v \Delta v + g(u, v) & x \in \Omega \quad t > 0 \end{aligned}$$

Initial conditions

$$u(x, 0) = u_0(x) \qquad v(x, 0) = v_0(x) \qquad x \in \Omega$$

Homogeneous Neumann boundary conditions

$$\nabla u \cdot \mathbf{n} = 0 \qquad \nabla v \cdot \mathbf{n} = 0 \qquad x \in \partial\Omega$$

- $f, g \in C^1(\mathbb{R}^2)$ describe local production, degradation and interactions at each spatial position
- $D_u, D_v > 0$ are the diffusion coefficients

Homogeneous steady state

HOMOGENEOUS EQUILIBRIUM

$$\begin{pmatrix} u(x, t) \\ v(x, t) \end{pmatrix} = \begin{pmatrix} \bar{u} \\ \bar{v} \end{pmatrix}$$

$$f(\bar{u}, \bar{v}) = 0 \quad g(\bar{u}, \bar{v}) = 0$$

LOCAL JACOBIAN

$$J(\bar{u}, \bar{v}) = \begin{pmatrix} f_u(\bar{u}, \bar{v}) & f_v(\bar{u}, \bar{v}) \\ g_u(\bar{u}, \bar{v}) & g_v(\bar{u}, \bar{v}) \end{pmatrix}$$

Both eigenvalues of $J(\bar{u}, \bar{v})$ have negative real parts

$$\text{tr } J(\bar{u}, \bar{v}) < 0 \quad \det J(\bar{u}, \bar{v}) > 0$$

$$u(x) = \bar{u}$$


$$v(x) = \bar{v}$$


the same concentrations at every position



- Stable without spatial variation**

The reaction kinetics restore every sufficiently small homogeneous perturbation.

Perturb both concentrations around the steady state

SMALL PERTURBATION

$$u(x, t) = \bar{u} + \eta(x, t)$$

$$v(x, t) = \bar{v} + \xi(x, t)$$



For infinitesimal perturbations, keep only first-order terms

Expand both reaction terms around $(\bar{u}, \bar{v})$

$$f(\bar{u} + \eta, \bar{v} + \xi) = f(\bar{u}, \bar{v}) + f_u(\bar{u}, \bar{v})\eta + f_v(\bar{u}, \bar{v})\xi + O(\|(\eta, \xi)\|^2)$$

$$g(\bar{u} + \eta, \bar{v} + \xi) = g(\bar{u}, \bar{v}) + g_u(\bar{u}, \bar{v})\eta + g_v(\bar{u}, \bar{v})\xi + O(\|(\eta, \xi)\|^2)$$

Since $(\bar{u}, \bar{v})$ is constant and $f(\bar{u}, \bar{v}) = g(\bar{u}, \bar{v}) = 0$, substitution gives

$$\partial_t \eta = D_u \Delta \eta + f_u(\bar{u}, \bar{v})\eta + f_v(\bar{u}, \bar{v})\xi + O(\|(\eta, \xi)\|^2)$$

$$\partial_t \xi = D_v \Delta \xi + g_u(\bar{u}, \bar{v})\eta + g_v(\bar{u}, \bar{v})\xi + O(\|(\eta, \xi)\|^2)$$

$$\partial_t \begin{pmatrix} \eta \\ \xi \end{pmatrix} = J(\bar{u}, \bar{v}) \begin{pmatrix} \eta \\ \xi \end{pmatrix} + \begin{pmatrix} D_u & 0 \\ 0 & D_v \end{pmatrix} \Delta \begin{pmatrix} \eta \\ \xi \end{pmatrix}$$

Spatial eigenfunctions separate the perturbation

Use the eigenfunctions introduced in the diffusion section

$$-\Delta \phi_k = \mu_k \phi_k \quad \nabla \phi_k \cdot n = 0 \quad \text{on } \partial\Omega$$

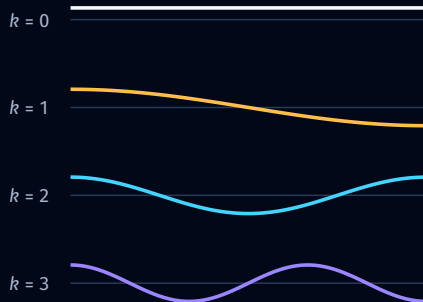
$$\begin{pmatrix} \eta(x, t) \\ \xi(x, t) \end{pmatrix} = \sum_{k=0}^{\infty} \begin{pmatrix} a_k(t) \\ b_k(t) \end{pmatrix} \phi_k(x)$$

- One spatial shape**

$\phi_k(x)$ determines where the perturbation is positive and negative.

- Two chemical amplitudes**

$a_k(t)$ and $b_k(t)$ determine how strongly that shape appears in each substance.



Each spatial shape will evolve independently after linearization.

Each mode is a two-dimensional stability problem

Combining the linearized perturbation equation with the eigenfunction expansion

$$\partial_t \begin{pmatrix} \eta \\ \xi \end{pmatrix} = J(\bar{u}, \bar{v}) \begin{pmatrix} \eta \\ \xi \end{pmatrix} + \begin{pmatrix} D_u & 0 \\ 0 & D_v \end{pmatrix} \Delta \begin{pmatrix} \eta \\ \xi \end{pmatrix} \quad + \quad \begin{pmatrix} \eta(x, t) \\ \xi(x, t) \end{pmatrix} = \sum_{k=0}^{\infty} \begin{pmatrix} a_k(t) \\ b_k(t) \end{pmatrix} \phi_k(x)$$



$$\sum_{k=0}^{\infty} \frac{d}{dt} \begin{pmatrix} a_k \\ b_k \end{pmatrix} \phi_k = J(\bar{u}, \bar{v}) \sum_{k=0}^{\infty} \begin{pmatrix} a_k \\ b_k \end{pmatrix} \phi_k + \begin{pmatrix} D_u & 0 \\ 0 & D_v \end{pmatrix} \sum_{k=0}^{\infty} \begin{pmatrix} a_k \\ b_k \end{pmatrix} \Delta \phi_k$$

Since $\Delta \phi_k = -\mu_k \phi_k$ and $\{\phi_k\}_{k=0}^{\infty}$ is an orthonormal basis, the coefficients of every ϕ_k must agree

$$\boxed{\frac{d}{dt} \begin{pmatrix} a_k \\ b_k \end{pmatrix} = \left(J(\bar{u}, \bar{v}) - \mu_k \begin{pmatrix} D_u & 0 \\ 0 & D_v \end{pmatrix} \right) \begin{pmatrix} a_k \\ b_k \end{pmatrix}}$$

Define the stability matrix of each mode

$$\frac{d}{dt} \begin{pmatrix} a_k \\ b_k \end{pmatrix} = \left(J(\bar{u}, \bar{v}) - \mu_k \begin{pmatrix} D_u & 0 \\ 0 & D_v \end{pmatrix} \right) \begin{pmatrix} a_k \\ b_k \end{pmatrix}$$

Define the stability matrix associated with the k th spatial mode

$$M_k = J(\bar{u}, \bar{v}) - \mu_k \begin{pmatrix} D_u & 0 \\ 0 & D_v \end{pmatrix} = \begin{pmatrix} f_u(\bar{u}, \bar{v}) - D_u \mu_k & f_v(\bar{u}, \bar{v}) \\ g_u(\bar{u}, \bar{v}) & g_v(\bar{u}, \bar{v}) - D_v \mu_k \end{pmatrix}$$

$$\frac{d}{dt} \begin{pmatrix} a_k \\ b_k \end{pmatrix} = M_k \begin{pmatrix} a_k \\ b_k \end{pmatrix}$$

Every spatial wavelength has its own 2×2 stability matrix.

The homogeneous mode remains locally stable

HOMOGENEOUS MODE

$$\phi_0(x) = \text{constant} \quad \mu_0 = 0$$

$$M_0 = J(\bar{u}, \bar{v})$$

the same perturbation everywhere

- Stable by assumption**

The local reaction analysis already guarantees decay of this mode.

NONCONSTANT MODES

$$\mu_k > 0 \quad k \geq 1$$

$$M_k = J(\bar{u}, \bar{v}) - \mu_k \begin{pmatrix} D_u & 0 \\ 0 & D_v \end{pmatrix}$$



a spatial pattern can change stability

- The Turing question**

Can any M_k acquire an eigenvalue with positive real part?

Diffusion makes the trace more negative

$$M_k = \begin{pmatrix} f_u(\bar{u}, \bar{v}) - D_u \mu_k & f_v(\bar{u}, \bar{v}) \\ g_u(\bar{u}, \bar{v}) & g_v(\bar{u}, \bar{v}) - D_v \mu_k \end{pmatrix}$$

$$\text{tr}(M_k) = \text{tr} J(\bar{u}, \bar{v}) - (D_u + D_v) \mu_k$$

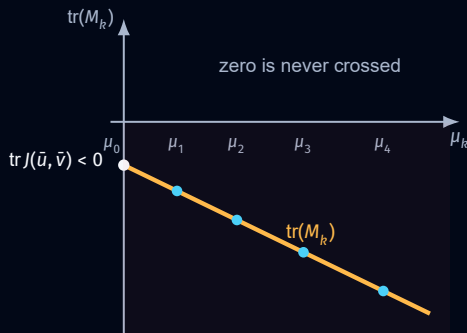
- The trace starts negative**

$\mu_0 = 0$ and $\text{tr}(M_0) = \text{tr} J(\bar{u}, \bar{v}) < 0$ because the homogeneous equilibrium is locally stable.

- Diffusion decreases it further**

For every $k \geq 1$, the term $-(D_u + D_v) \mu_k$ is strictly negative.

Therefore a Turing instability cannot occur through the trace becoming positive.



The determinant contains the instability mechanism

Compute the determinant of the mode matrix

$$\begin{aligned}
 \det(M_k) &= \det \begin{pmatrix} f_u(\bar{u}, \bar{v}) - D_u \mu_k & f_v(\bar{u}, \bar{v}) \\ g_u(\bar{u}, \bar{v}) & g_v(\bar{u}, \bar{v}) - D_v \mu_k \end{pmatrix} \\
 &= (f_u(\bar{u}, \bar{v}) - D_u \mu_k)(g_v(\bar{u}, \bar{v}) - D_v \mu_k) - f_v(\bar{u}, \bar{v})g_u(\bar{u}, \bar{v}) \\
 &= \det J(\bar{u}, \bar{v}) - (D_v f_u(\bar{u}, \bar{v}) + D_u g_v(\bar{u}, \bar{v}))\mu_k + D_u D_v \mu_k^2
 \end{aligned}$$

$$\det(M_k) = \det J(\bar{u}, \bar{v}) - A\mu_k + D_u D_v \mu_k^2$$

$$A = D_v f_u(\bar{u}, \bar{v}) + D_u g_v(\bar{u}, \bar{v})$$

- **Positive at $\mu_0 = 0$**
 $\det(M_0) = \det J(\bar{u}, \bar{v}) > 0$ follows from local stability.
- **The linear term can decrease it**
 Unequal diffusion weights f_u and g_v differently inside A .
- **Very short waves are stable**
 $D_u D_v \mu_k^2$ dominates for sufficiently large k .

Turing instability condition

$$\det(M_k) = \det J(\bar{u}, \bar{v}) - A\mu_k + D_u D_v \mu_k^2$$

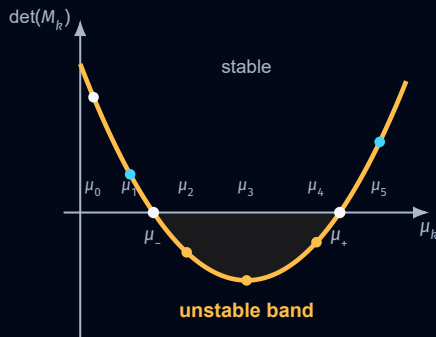
- $\det(M_k) > 0$
The k th spatial mode is stable.
- $\det(M_k) < 0$
The k th spatial mode is unstable.

$$\det(M_k) < 0 \iff \mu_- < \mu_k < \mu_+$$

A negative interval exists when

$$A > 0 \quad A^2 > 4D_u D_v \det J(\bar{u}, \bar{v})$$

$$\mu_{\pm} = \frac{A \pm \sqrt{A^2 - 4D_u D_v \det J(\bar{u}, \bar{v})}}{2D_u D_v}$$



Turing Instability Theorem

$$\begin{aligned}\partial_t u &= D_u \Delta u + f(u, v) \\ \partial_t v &= D_v \Delta v + g(u, v)\end{aligned}$$

THEOREM

If the system is stable in the absence of diffusion, and u satisfies the autocatalysis condition, i.e. $f_u(\bar{u}, \bar{v}) > 0$, diffusion D_u is sufficiently small and diffusion D_v is sufficiently large, then diffusion destabilizes the homogeneous steady state $(\bar{u}, \bar{v})$.

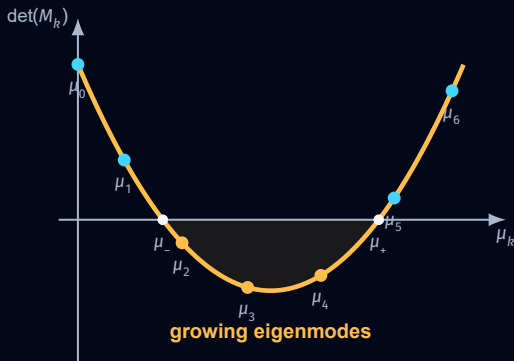
03 TURING INSTABILITY

Node selection

The unstable band must intersect the discrete spatial spectrum allowed by the tissue geometry



Only eigenvalues inside the unstable band can grow



- Only selected modes grow**

The unstable band is continuous, but the tissue supplies only discrete eigenvalues.

$$\det(M_k) < 0$$

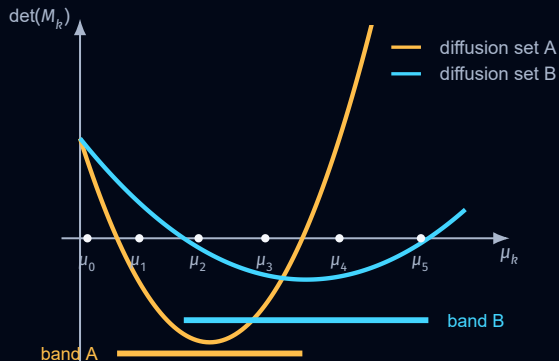


Exists a positive eigenvalue $\lambda_{k,+}$



Spatial mode k grows at rate $e^{\lambda_{k,+} t}$

Diffusion changes which nodes are unstable



$$\det(M_k) = \det J(\bar{u}, \bar{v}) - A\mu_k + D_u D_v \mu_k^2$$

- **The band moves**

Changing D_u and D_v changes its location and width.

- **The selected modes change**

A different set of the fixed eigenvalues μ_k can become unstable.

Diffusion therefore controls the preferred spatial scale.

Diffusion and domain size

Let $\tilde{\Omega}$ be a fixed reference domain and scale it by $L > 0$

$$\Omega_L = L\tilde{\Omega} \quad x = L\tilde{x}$$

Under the change of variables $x = L\tilde{x}$, the system is equivalently written on the reference domain as

$$\partial_t \tilde{u} = \frac{D_u}{L^2} \Delta_{\tilde{x}} \tilde{u} + f(\tilde{u}, \tilde{v})$$

$$\partial_t \tilde{v} = \frac{D_v}{L^2} \Delta_{\tilde{x}} \tilde{v} + g(\tilde{u}, \tilde{v})$$

$$L\tilde{\Omega} \text{ with } (D_u, D_v) \Leftrightarrow \tilde{\Omega} \text{ with } \left(\frac{D_u}{L^2}, \frac{D_v}{L^2} \right)$$

small domain



large domain



increasing L compresses the spatial spectrum

- Important distinction**

Changing the ratio D_v/D_u is not equivalent to changing the domain size.

A larger domain can support more peaks

SMALL DOMAIN

homogeneous steady state with a small perturbation



one peak



LARGER DOMAIN

homogeneous steady state with a small perturbation



multiple peaks



- **Linear level**

Turing analysis identifies which modes lie in the unstable band and can grow

- **Pattern formation requires nonlinearity**

Nonlinear interactions select and saturate the emerging spatial pattern

- **Useful but incomplete prediction**

Unstable eigenmodes can be useful for predicting the pattern, but they do not provide a complete explanation

LECTURE MAP

The road from Hydra to pattern formation

From biological observations to mathematical mechanisms.

01 The biological puzzle

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04 Gierer–Meinhardt model

05 Growth, wounds & long-range dynamics

06 Mechanochemical pattern formation



Return to the Gierer–Meinhardt model

Let $\Omega \subset \mathbb{R}^n$ be a bounded domain

$$\partial_t u = D_u \Delta u + \frac{au^2}{K + v} - \mu_u u$$

$$\partial_t v = D_v \Delta v + bu^2 - \mu_v v$$

Initial conditions

$$u(x, 0) = u_0(x) \quad v(x, 0) = v_0(x) \quad x \in \Omega$$

Homogeneous Neumann boundary conditions

$$\nabla u \cdot \mathbf{n} = 0 \quad \nabla v \cdot \mathbf{n} = 0 \quad x \in \partial\Omega$$

u activator concentration

v inhibitor concentration

D_u, D_v diffusion coefficients

a activator self-amplification

K inhibition scale

b inhibitor production

μ_u, μ_v degradation rates

Homogeneous states

Which constant concentrations can persist in time?

Local stability

Which states recover after a spatially uniform perturbation?

Diffusion-driven instability

Can diffusion destabilize a locally stable active state?

04 GIERER-MEINHARDT MODEL

Steady states

The geometry of the local reactions reveals an activation threshold and two competing stable outcomes



Reaction system

Looking for steady states and analysing stability

$$\begin{aligned}\partial_t u &= f(u, v) = \frac{au^2}{K + v} - \mu_u u \\ \partial_t v &= g(u, v) = bu^2 - \mu_v v\end{aligned}$$

Without diffusion every spatial position follows the same two-variable reaction system

- **Steady states**

Find constant concentrations $(\bar{u}, \bar{v})$ for which $f(\bar{u}, \bar{v}) = 0$ and $g(\bar{u}, \bar{v}) = 0$

- **Local stability**

Linearize the reaction system around each steady state and inspect the eigenvalues of the Jacobian

Steady states

A homogeneous steady state $(\bar{u}, \bar{v})$ satisfies

$$\begin{aligned} 0 &= \frac{a\bar{u}^2}{K + \bar{v}} - \mu_u \bar{u} \\ 0 &= b\bar{u}^2 - \mu_v \bar{v} \end{aligned}$$

The second equation gives

$$\bar{v} = \frac{b}{\mu_v} \bar{u}^2$$

Substitution into the first equation gives

$$0 = \frac{a\bar{u}^2}{K + \frac{b}{\mu_v} \bar{u}^2} - \mu_u \bar{u}$$

After clearing the denominator

$$\bar{u} \left(\frac{b}{\mu_v} \bar{u}^2 - \frac{a}{\mu_u} \bar{u} + K \right) = 0$$

A cubic equation determines all homogeneous steady states

Steady states

$$\bar{u} = 0 \quad \Rightarrow \quad (\bar{u}, \bar{v}) = (0, 0)$$

$$\bar{u} > 0 \quad \Rightarrow \quad \frac{b}{\mu_v} \bar{u}^2 - \frac{a}{\mu_u} \bar{u} + K = 0$$

TWO POSITIVE STEADY STATES IF $a^2 \mu_v > 4bK\mu_u^2$

$$u_{\pm} = \frac{\mu_v}{2b} \left[\frac{a}{\mu_u} \pm \sqrt{\left(\frac{a}{\mu_u} \right)^2 - \frac{4bK}{\mu_v}} \right]$$

$$v_{\pm} = \frac{b}{\mu_v} u_{\pm}^2$$

- **Lower state** (u_-, v_-)

The smaller positive root will become the activation threshold

- **Upper state** (u_+, v_+)

The larger positive root is the candidate active state

The nonlinear reaction terms can create two distinct positive concentration levels

The nullclines

ACTIVATOR NULLCLINE

$$f(u, v) = \frac{au^2}{K + v} - \mu_u u = 0$$

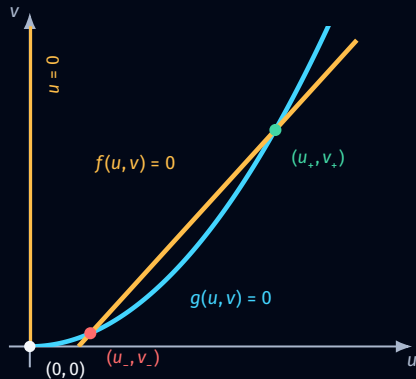
$$\iff u = 0 \quad \text{or} \quad v = \frac{a}{\mu_u} u - K$$

INHIBITOR NULLCLINE

$$g(u, v) = bu^2 - \mu_v v = 0$$

$$\iff v = \frac{b}{\mu_v} u^2$$

Positive equilibria are intersections of a straight line and a parabola



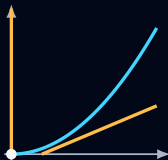
Possible steady states

EXISTENCE CONDITION

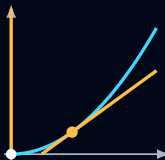
$$\left(\frac{a}{\mu_u}\right)^2 > \frac{4bK}{\mu_v}$$

At equality, the line is tangent to the parabola and the two positive equilibria merge

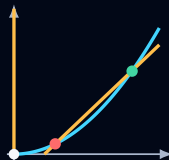
NO POSITIVE STATE

1 steady state
(0, 0)

TANGENCY

2 steady states
(0, 0) (u_c, v_c)

TWO POSITIVE STATES

3 steady states
(0, 0) (u_-, v_-) (u_+, v_+)

Strong enough autocatalysis creates the zero state and two positive states separated by a threshold

04 GIERER-MEINHARDT MODEL

Local stability

We first analyse stability in the local reaction system without diffusion and identify the activation threshold



The zero steady state is always stable

$$f(u, v) = \frac{au^2}{K + v} - \mu_u u \quad g(u, v) = bu^2 - \mu_v v$$

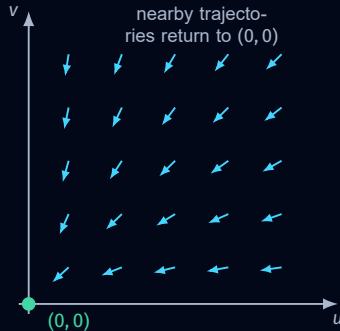
JACOBIAN

$$J(u, v) = \begin{pmatrix} f_u(u, v) & f_v(u, v) \\ g_u(u, v) & g_v(u, v) \end{pmatrix} = \begin{pmatrix} \frac{2au}{K + v} - \mu_u & -\frac{au^2}{(K + v)^2} \\ 2bu & -\mu_v \end{pmatrix}$$

$$J(0, 0) = \begin{pmatrix} -\mu_u & 0 \\ 0 & -\mu_v \end{pmatrix}, \quad \begin{aligned} \lambda_1 &= -\mu_u < 0 \\ \lambda_2 &= -\mu_v < 0 \end{aligned}$$

• Asymptotically stable

Near $u = 0$ quadratic activator production is dominated by linear degradation



Jacobian at the positive steady states

At either positive steady state $(u_{\pm}, v_{\pm})$ the activator equation gives

$$f(u_{\pm}, v_{\pm}) = \frac{au_{\pm}^2}{K + v_{\pm}} - \mu_u u_{\pm} = 0 \quad \Rightarrow \quad \frac{au_{\pm}}{K + v_{\pm}} = \mu_u$$

ACTIVATOR DERIVATIVES

$$f_u(u_{\pm}, v_{\pm}) = \frac{2au_{\pm}}{K + v_{\pm}} - \mu_u = \mu_u$$

$$f_v(u_{\pm}, v_{\pm}) = -\frac{au_{\pm}^2}{(K + v_{\pm})^2} = -\frac{\mu_u^2}{a}$$

INHIBITOR DERIVATIVES

$$g_u(u_{\pm}, v_{\pm}) = 2bu_{\pm}$$

$$g_v(u_{\pm}, v_{\pm}) = -\mu_v$$

$$J(u_{\pm}, v_{\pm}) = \begin{pmatrix} \mu_u & -\mu_u^2/a \\ 2bu_{\pm} & -\mu_v \end{pmatrix}$$

The two positive steady states differ only through the entry $2bu_{\pm}$

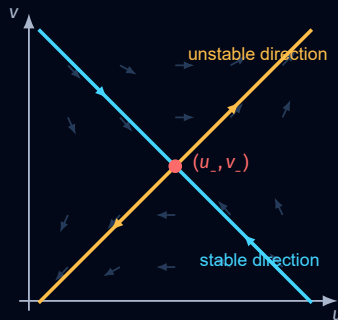
The lower positive steady state is a saddle

$$\begin{aligned}\operatorname{tr} J(u_{\pm}, v_{\pm}) &= \mu_u - \mu_v \\ \det J(u_{\pm}, v_{\pm}) &= \mu_u \left(\frac{2b\mu_u}{a} u_{\pm} - \mu_v \right)\end{aligned}$$

The two positive roots lie on opposite sides of their midpoint

$$u_- < \frac{a\mu_v}{2b\mu_u} < u_+$$

$$\boxed{\det J(u_-, v_-) < 0} \quad \Rightarrow \quad \lambda_1 \lambda_2 < 0$$



• Saddle

One eigenvalue is negative and the other is positive

The upper positive steady state can be stable

$$u_+ > \frac{a\mu_v}{2b\mu_u}$$

$$\det J(u_+, v_+) > 0$$

$$\operatorname{tr} J(u_+, v_+) = \mu_u - \mu_v$$

$$\boxed{\mu_u < \mu_v} \Rightarrow \operatorname{tr} J(u_+, v_+) < 0$$

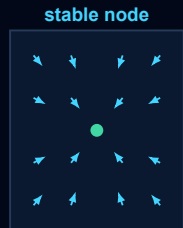
$$\Rightarrow \operatorname{Re} \lambda_1 < 0 \quad \operatorname{Re} \lambda_2 < 0$$

The discriminant determines the local geometry

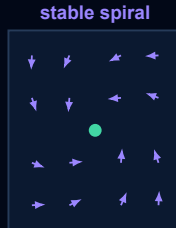
$$D_+ = (\operatorname{tr} J(u_+, v_+))^2 - 4 \det J(u_+, v_+)$$

$$D_+ \geq 0 \Rightarrow \text{stable node}$$

$$D_+ < 0 \Rightarrow \text{stable spiral}$$

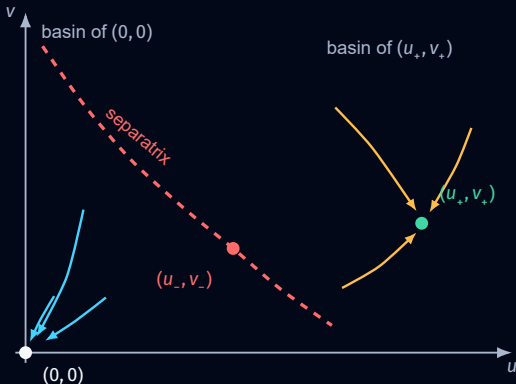


(u_+, v_+)



(u_+, v_+)

The saddle separates the two stable outcomes



- **Small perturbations**
States below the threshold return to $(0, 0)$
- **Activation threshold**
The stable manifold of (u_-, v_-) forms the basin boundary
- **Large perturbations**
States beyond the threshold approach (u_+, v_+)

The final state depends on whether the perturbation crosses the nonlinear threshold

Turing instability

Diffusion acts on the stable positive state
and selects a band of growing
spatial eigenmodes



Choose the homogeneous steady state

CONSTANT POSITIVE STATE

$$\begin{pmatrix} u(x, t) \\ v(x, t) \end{pmatrix} = \begin{pmatrix} u_+ \\ v_+ \end{pmatrix} \quad x \in \Omega$$

The same concentrations occur at every spatial position

STABLE WITHOUT DIFFUSION

$$\begin{aligned} \operatorname{tr} J(u_+, v_+) &= \mu_u - \mu_v < 0 \\ \det J(u_+, v_+) &> 0 \end{aligned}$$

$$\mu_u < \mu_v$$

$$J(u_+, v_+) = \begin{pmatrix} \mu_u & -\mu_u^2/a \\ 2bu_+ & -\mu_v \end{pmatrix}$$

The general Turing theorem applies to GM

$$J(u_+, v_+) = \begin{pmatrix} \mu_u & -\mu_u^2/a \\ 2bu_+ & -\mu_v \end{pmatrix} \quad f_u(u_+, v_+) = \mu_u > 0$$

THEOREM

If a two-variable reaction system is stable without diffusion and $f_u(\bar{u}, \bar{v}) > 0$ then D_u can be chosen sufficiently small and D_v sufficiently large so that diffusion destabilizes the homogeneous steady state

Suitable diffusion coefficients can destabilize the stable positive GM steady state

Each spatial mode gives a two-dimensional system

SPATIAL EIGENFUNCTION

$$-\Delta \phi_k = \mu_k \phi_k$$

$$\mu_k \longrightarrow \infty \quad k \longrightarrow \infty$$

AMPLITUDE EQUATION

$$\frac{d}{dt} \begin{pmatrix} \alpha_k \\ \beta_k \end{pmatrix} = M_k \begin{pmatrix} \alpha_k \\ \beta_k \end{pmatrix}$$

Perturb both concentrations in the same spatial mode

$$u(x, t) = u_+ + \alpha_k(t) \phi_k(x)$$

$$v(x, t) = v_+ + \beta_k(t) \phi_k(x)$$

$$M_k = J(u_+, v_+) - \mu_k \begin{pmatrix} D_u & 0 \\ 0 & D_v \end{pmatrix} = \begin{pmatrix} \mu_u - D_u \mu_k & -\mu_u^2/a \\ 2bu_+ & -\mu_v - D_v \mu_k \end{pmatrix}$$

Every eigenmode has its own explicit 2×2 stability matrix

Exact diffusion conditions for Turing instability

$$M_k = \begin{pmatrix} \mu_u - D_u \mu_k & -\mu_u^2/a \\ 2bu_+ & -\mu_v - D_v \mu_k \end{pmatrix}$$

$$\det(M_k) = \det J(u_+, v_+) - A_{\text{GM}} \mu_k + D_u D_v \mu_k^2$$

$$A_{\text{GM}} = D_v \mu_u - D_u \mu_v$$

- The parabola must initially decrease**

$A_{\text{GM}} > 0$ is equivalent to faster inhibitor diffusion

$$A_{\text{GM}} > 0 \iff \boxed{\frac{D_v}{D_u} > \frac{\mu_v}{\mu_u}}$$

- The minimum must cross zero**

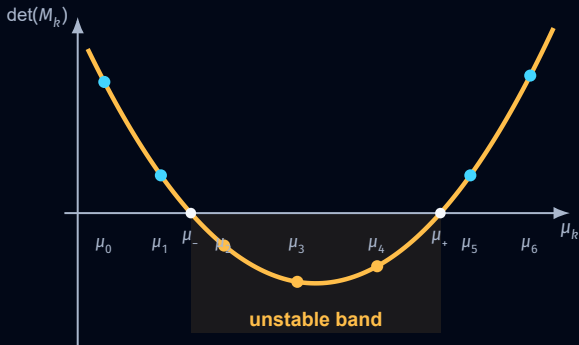
The quadratic must have two distinct positive roots

$$A_{\text{GM}}^2 > 4D_u D_v \det J(u_+, v_+)$$

$$\mu_{\pm} = \frac{A_{\text{GM}} \pm \sqrt{A_{\text{GM}}^2 - 4D_u D_v \det J(u_+, v_+)}}{2D_u D_v}$$

$$\det(M_k) < 0 \iff \mu_- < \mu_k < \mu_+$$

Only GM eigenmodes inside the unstable band can grow



- Unstable modes**

Only discrete eigenvalues μ_k between μ_- and μ_+ can grow

- Stable modes**

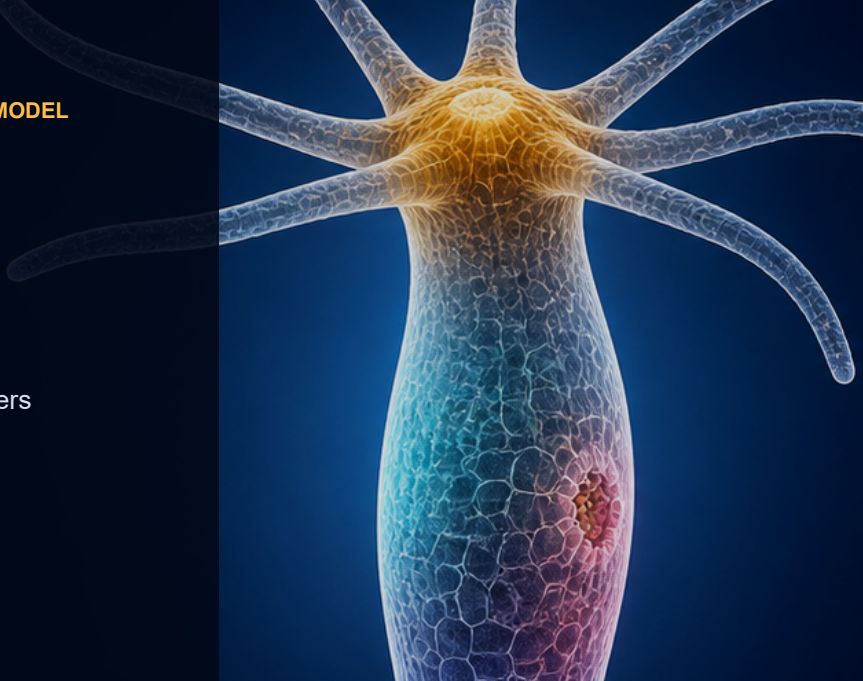
Modes outside the negative determinant interval decay

$$\det(M_k) < 0 \Rightarrow \lambda_{k,+} > 0 \Rightarrow (\alpha_k(t), \beta_k(t)) \sim e^{\lambda_{k,+} t}$$

04 GIERER-MEINHARDT MODEL

Example

Certain set of parameters



Example

Let $\Omega = [0, 1] \subset \mathbb{R}$ be an interval

$$\partial_t u = 0.01 \Delta u + \frac{1.5u^2}{1+v} - 0.5u$$

$$\partial_t v = \Delta v + 2u^2 - v$$

PARAMETER VALUES

$$D_u = 0.01$$

$$D_v = 1$$

$$a = 1.5$$

$$b = 2$$

$$\mu_u = 0.5$$

$$\mu_v = 1$$

$$K = 1$$

- Initial conditions**

$$u(x, 0) = u_0(x) \text{ and } v(x, 0) = v_0(x)$$

- No-flux boundaries**

$$\partial_x u = \partial_x v = 0 \text{ at } x = 0 \text{ and } x = 1$$

Three steady states

Constant steady states solve

$$0 = \frac{1.5\bar{u}^2}{1 + \bar{v}} - 0.5\bar{u}$$

$$0 = 2\bar{u}^2 - \bar{v}$$

$$\bar{v} = 2\bar{u}^2$$

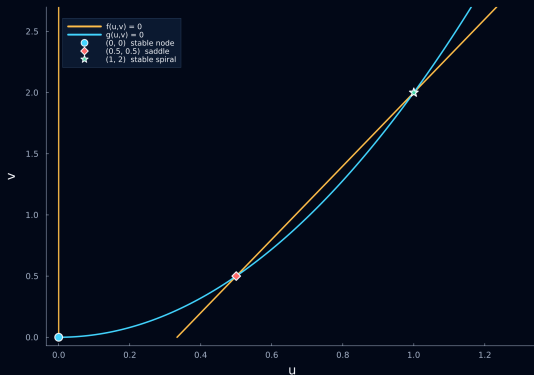
$$\bar{u} (2\bar{u}^2 - 3\bar{u} + 1) = \bar{u} (2\bar{u} - 1)(\bar{u} - 1) = 0$$

$$(\bar{u}, \bar{v}) = (0, 0)$$

$$(\bar{u}, \bar{v}) = (0.5, 0.5)$$

$$(\bar{u}, \bar{v}) = (1, 2)$$

Nullclines



- **Activator nullcline**
 $u = 0$ or $v = 3u - 1$
- **Inhibitor nullcline**
 $v = 2u^2$
- **Three intersections**
 Every intersection is a steady state of the local reaction system

Steady states

Evaluate the Jacobian and its eigenvalues at each steady state

$$(\bar{u}, \bar{v}) = (0, 0)$$

$$\lambda_1 = -1$$

$$\lambda_2 = -0.5$$

- **Stable node**

Both eigenvalues are negative

$$(\bar{u}, \bar{v}) = (0.5, 0.5)$$

$$\lambda_- = -0.72871$$

$$\lambda_+ = 0.22871$$

- **Saddle**

One eigenvalue is positive

$$(\bar{u}, \bar{v}) = (1, 2)$$

$$\lambda_{\pm} = -0.25 \pm 0.32275i$$

- **Stable spiral**

Both real parts are negative

Only the middle steady state has a positive local eigenvalue

Diffusion destabilizes the upper steady state

Linearization at the stable steady state (1, 2)

$$J(1, 2) = \begin{pmatrix} 0.5 & -1/6 \\ 4 & -1 \end{pmatrix}$$

$$\text{tr } J(1, 2) = -0.5$$

$$\det J(1, 2) = \frac{1}{6}$$

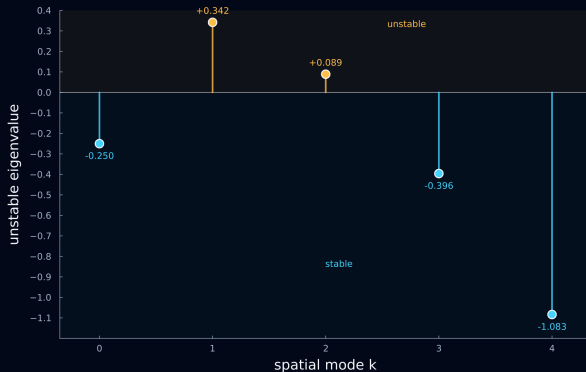
$$M_k = \begin{pmatrix} 0.5 - 0.01\mu_k & -1/6 \\ 4 & -1 - \mu_k \end{pmatrix}$$

$$\det(M_k) = \frac{1}{6} - 0.49\mu_k + 0.01\mu_k^2$$

$$\det(M_k) < 0 \iff 0.34253 < \mu_k < 48.65747$$

Modes whose Laplacian eigenvalues lie inside this interval are unstable

Unstable eigenmodes



$$D_u = 0.01 \quad D_v = 1 \quad L = 1$$

- **Dominant mode $k = 1$**
 $\lambda_{1,+} = 0.34184$
- **Slower mode $k = 2$**
 $\lambda_{2,+} = 0.08878$
- **Stable neighbours**
 $k = 0, 3, \dots$ decay

The first five spatial eigenmodes on $\Omega = [0, 1]$

Steady states of the local reaction system

$$(0, 0) \quad (0.5, 0.5) \quad (1, 2)$$

$$(\bar{u}, \bar{v}) = (1, 2)$$

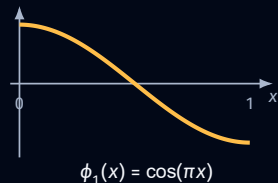
LINEARIZATION USED BELOW

$$\phi_k(x) = \cos(k\pi x)$$

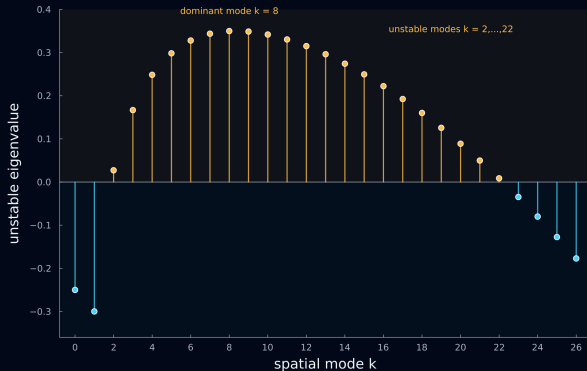
k	$\phi_k(x)$	ρ_k	behaviour
1	$\cos(\pi x)$	+0.34184	unstable
2	$\cos(2\pi x)$	+0.08878	unstable
3	$\cos(3\pi x)$	-0.39572	stable
4	$\cos(4\pi x)$	-1.08336	stable
5	$\cos(5\pi x)$	-1.97011	stable

$$\rho_k = \max_{\lambda \in \sigma(M_k)} \operatorname{Re} \lambda$$

DOMINANT UNSTABLE MODE



Unstable eigenmodes on bigger domain $\Omega = [0, 10]$



$$D_u = 0.01, \quad D_v = 1, \quad L = 10$$

- **Many unstable modes**
 $k = 2, \dots, 22$
- **Dominant mode $k = 8$**
 $\lambda_{8,+} = 0.34988$
- **Stable neighbours**
 $k = 0, 1, 23, \dots$ decay

The longer domain places more discrete eigenvalues inside the same unstable band

Selected spatial eigenmodes on $\Omega = [0, 10]$

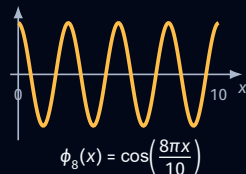
 $(0, 0)$ $(0.5, 0.5)$ $(1, 2)$ $(\bar{u}, \bar{v}) = (1, 2)$

$$\phi_k(x) = \cos\left(\frac{k\pi x}{10}\right)$$

$$\rho_k = \max_{\lambda \in \sigma(M_k)} \operatorname{Re} \lambda$$

k	$\phi_k(x)$	ρ_k	behaviour
1	$\cos(\pi x/10)$	-0.29984	stable
2	$\cos(2\pi x/10)$	+0.02724	unstable
6	$\cos(6\pi x/10)$	+0.32788	unstable
7	$\cos(7\pi x/10)$	+0.34376	unstable
8	$\cos(8\pi x/10)$	+0.34988	unstable
9	$\cos(9\pi x/10)$	+0.34870	unstable
10	$\cos(10\pi x/10)$	+0.34184	unstable
21	$\cos(21\pi x/10)$	+0.04979	unstable
22	$\cos(22\pi x/10)$	+0.00864	unstable
23	$\cos(23\pi x/10)$	-0.03464	stable

DOMINANT UNSTABLE MODE

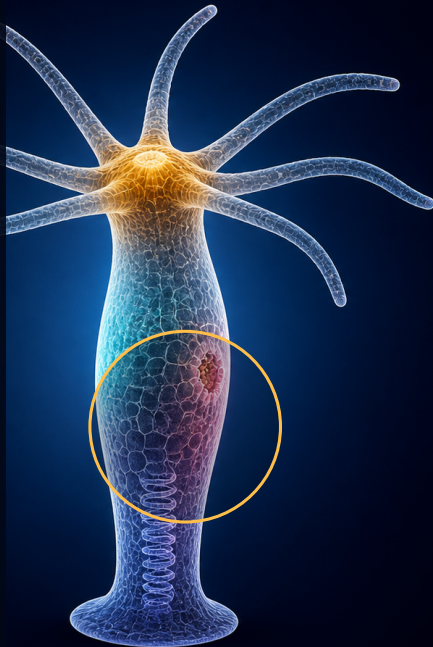


LECTURE MAP

The road from Hydra to pattern formation

From biological observations to mathematical mechanisms.

- 01 The biological puzzle
- 02 Reaction–diffusion approach
- 03 Turing instability
- 04 Gierer–Meinhardt model
- 05 Growth, wounds & long-range dynamics**
- 06 Mechanochemical pattern formation



What do we mean by a pattern?

Return to the Gierer–Meinhardt system

$$\partial_t u = D_u \Delta u + \frac{au^2}{K+v} - \mu_u u$$

$$\partial_t v = D_v \Delta v + bu^2 - \mu_v v$$

with homogeneous Neumann boundary conditions on Ω

STEADY STATE

$$0 = D_u \Delta \bar{u} + \frac{a\bar{u}^2}{K+\bar{v}} - \mu_u \bar{u}$$

$$0 = D_v \Delta \bar{v} + b\bar{u}^2 - \mu_v \bar{v}$$

- Spatial pattern**

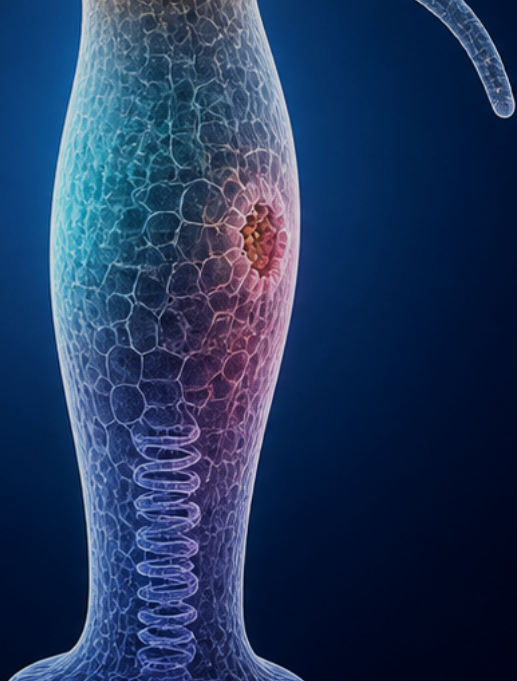
A stable nonconstant steady state $(\bar{u}(x), \bar{v}(x))$

For nonlinear reaction–diffusion systems, a complete characterization of all steady states is generally not known

05 GROWTH, WOUNDS & LONG-RANGE DYNAMICS

Growth and division

Growth and division change both the geometry and the initial data of the nonlinear problem



Growth makes the domain part of the dynamics

TIME-DEPENDENT DOMAIN

$$\Omega(t) = [0, L(t)]$$

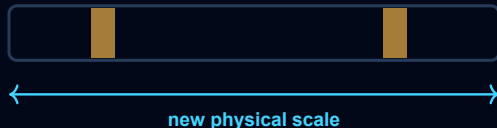
$$L'(t) > 0$$

- Changing geometry**

Growth changes physical distances and the spatial structures that fit inside the domain

- Nonlinear question**

Does the current pattern persist, reorganize, or generate additional peaks

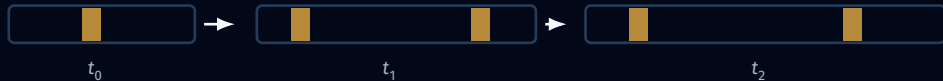


The concentrations evolve while the domain supporting them changes

Growth can transform one stable pattern into another

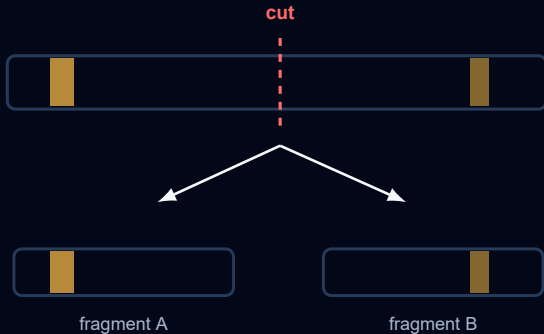
a new spatial structure can form as the domain grows

the simulation follows the full nonlinear solution in the physical coordinate



Growth can reorganize an existing pattern instead of merely stretching it

Division creates two new initial-value problems

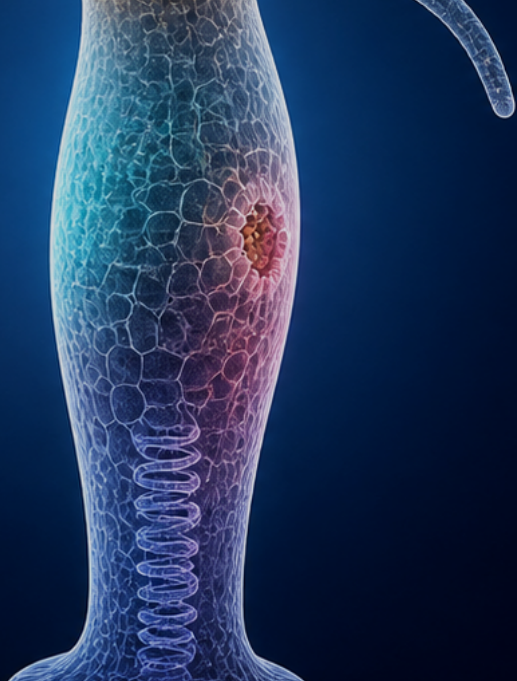


- **Inherited profiles**
Each fragment receives the restriction of the pre-cut profiles of u and v
- **New boundaries**
No-flux conditions are imposed at the newly created ends
- **Independent evolution**
After division the fragments solve two separate nonlinear problems

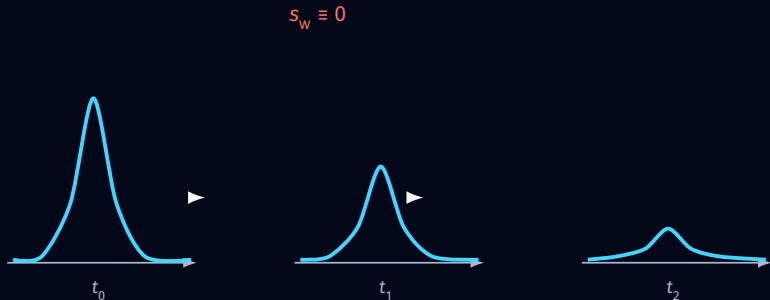
05 GROWTH, WOUNDS & LONG-RANGE DYNAMICS

Role of the wound

Regeneration requires a localized wound signal strong enough to cross the nonlinear activation threshold



A cut alone does not guarantee regeneration



A new boundary does not by itself move the system into the basin of a nonzero pattern

- **Simulation outcome**

Without wound input the headless fragment can converge to $(0, 0)$

- **Nonlinear explanation**

The inherited profile remains below the activation threshold and inside the basin of $(0, 0)$

A wound provides a localized transient signal

ACTIVATOR EQUATION WITH WOUND INPUT

$$\partial_t u = D_u \Delta u + \frac{au^2}{K+u} - \mu_u u + s_w(x, t)$$

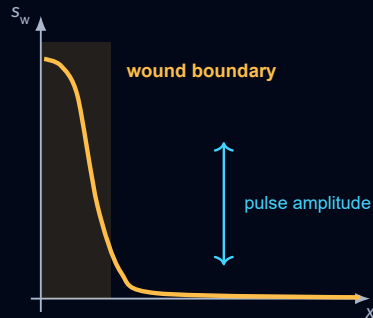
$$\text{supp } s_w \subset [0, \ell_w] \times [0, \tau_w]$$

- Localized in space**

The perturbation acts near the newly created wound boundary

- Transient in time**

The input disappears after a finite activation period



- Threshold crossing**

Sufficient amplitude and duration move the local state across the activation threshold

A transient wound can create a persistent organizer

the wound input disappears while the newly initiated peak persists

after threshold crossing, the autonomous GM dynamics can approach a stable nonconstant steady state



The wound initiates regeneration while the nonlinear feedback maintains the resulting organizer

The simulations expose a missing biological mechanism

WHAT THE SIMULATIONS DEMONSTRATE

- **Several stable patterns**

The full nonlinear system can support different steady spatial profiles

- **Geometry-dependent outcomes**

Growth and division change the domain and the initial-value problem

- **Wound-dependent regeneration**

A transient localized input can initiate a persistent organizer

WHAT THE MODEL STILL ASSUMES

- **Long-range molecular inhibition**

The classical mechanism requires an inhibitor that spreads farther than the activator

- **Uncertain biological identity**

Candidate activators exist, but the corresponding rapidly spreading inhibitor remains unclear

Could mechanics supply the missing inhibition?

This motivates a mechanochemical model in which tissue deformation participates in pattern formation

LECTURE MAP

The road from Hydra to pattern formation

From biological observations to mathematical mechanisms.

01 The biological puzzle

02 Reaction–diffusion approach

03 Turing instability

04 Gierer–Meinhardt model

05 Growth, wounds & long-range dynamics

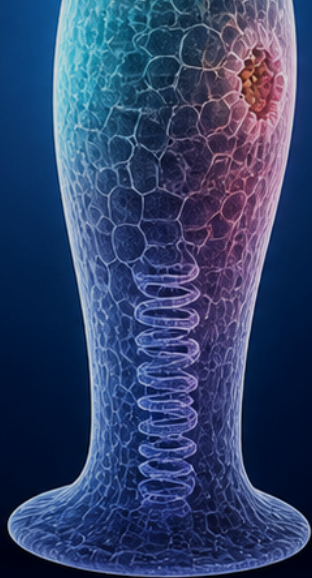
06 Mechanochemical pattern formation



06 MECHANOCHEMICAL PATTERN FORMATION

Why mechanics?

The classical inhibitor is not identified and mechanics offers an alternative long-range coupling



Return to the Gierer–Meinhardt model

The chemical mechanism uses two interacting substances

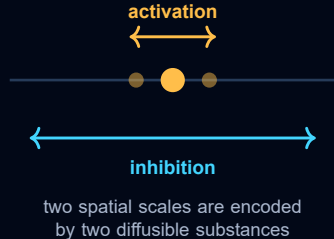
$$\begin{aligned}\partial_t u &= D_u \Delta u + \frac{au^2}{K+v} - \mu_u u \\ \partial_t v &= D_v \Delta v + bu^2 - \mu_v v\end{aligned}$$

- Local activation**

The activator promotes its own production over a short range

- Long-range inhibition**

A second substance suppresses activation over a larger spatial scale



The activator has candidates but the inhibitor remains unclear

ACTIVATOR CANDIDATE

Wnt3

Stable localized Wnt3 expression marks the emerging head organizer

MISSING COMPONENT

?

No biochemical species has been established as the required fast long-range inhibitor

The mathematical inhibitor has a clear role but its biological identity is unresolved

Long-range inhibition need not be chemical

- **Chemical transport**

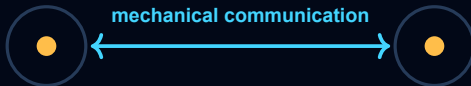
A molecule would have to transmit inhibition across the regenerating tissue

- **Mechanical coupling**

Stress and deformation can couple distant parts of a closed tissue without a second inhibitor

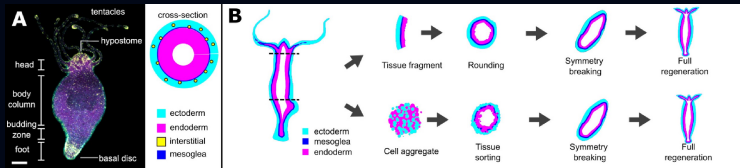
- **Modelling question**

Can one morphogen and tissue mechanics reproduce local activation and long-range inhibition



a global constraint can transmit local changes throughout the tissue

Symmetry breaking occurs in a regenerating spheroid



- **Initially symmetric**
A tissue fragment rounds into a hollow spheroid
- **Symmetry is broken**
One region becomes the future head organizer
- **A complete axis emerges**
The localized organizer directs whole-body regeneration

The central question

Can a local morphogen change the mechanics of the entire spheroid

**Can mechanics replace the diffusible
inhibitor**

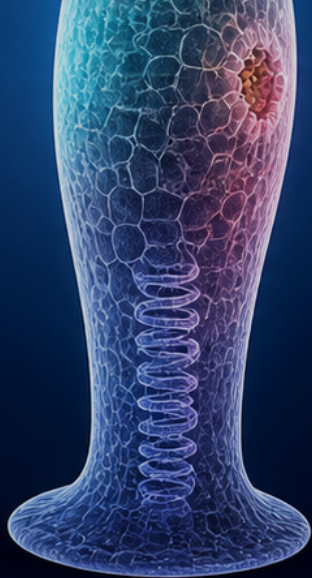


**Build the simplest mechanochemical model that can answer
this question**

06 MECHANOCHEMICAL PATTERN FORMATION

Mechanics during regeneration

Swelling and rupture expose a two-way coupling between morphogen signalling and mechanics



Regenerating spheroids undergo mechanical oscillations

REGENERATING HYDRA SPHEROID



observe gradual swelling followed by rapid collapse

One oscillation is a swelling and rupture cycle

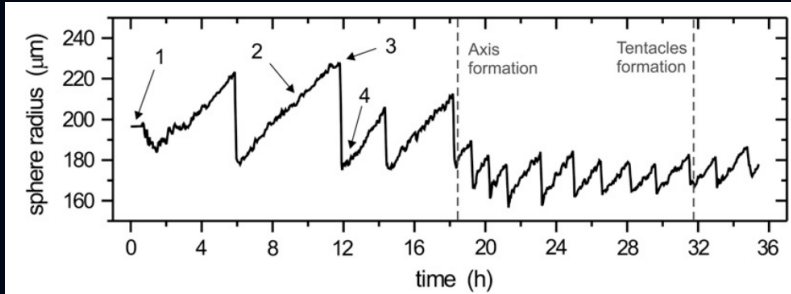
water enters → tissue stretches → tissue ruptures



the tissue deflates heals and the cycle begins again

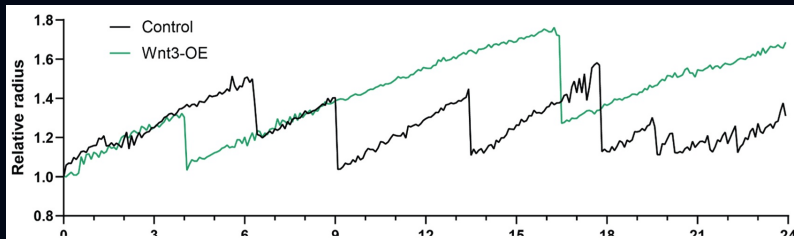
Osmotic pressure repeatedly converts fluid transport into tissue stretching

The spheroid radius records repeated rupture events



- **Slow increase**
Water influx gradually increases the spheroid radius
- **Rapid decrease**
Rupture releases fluid and abruptly reduces the radius
- **Changing dynamics**
The oscillation changes as the body axis and tentacles form

Wnt3 changes the oscillatory dynamics



- **Control**

Repeated rupture limits the gradual increase in radius

- **Wnt3 overexpression**

The timing and amplitude of rupture events are altered

Morphogen signalling and tissue-scale mechanics cannot be treated independently

Experiments suggest a two-way mechanochemical coupling

TISSUE STRETCHING

⇓ Wnt3 production

Stretching correlates with organizer formation and Wnt3 expression

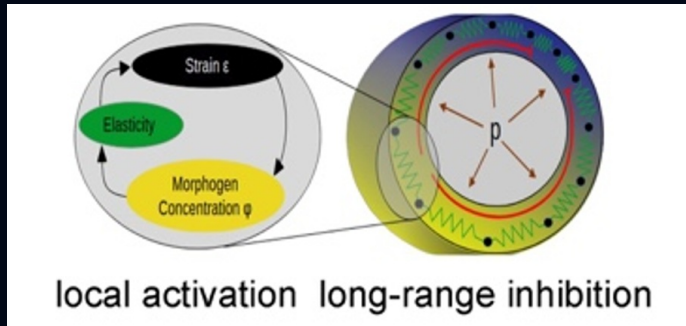
MORPHOGEN

⇓ Tissue elasticity

Morphogen signalling changes how strongly the tissue resists deformation

Each side of the feedback modifies the other

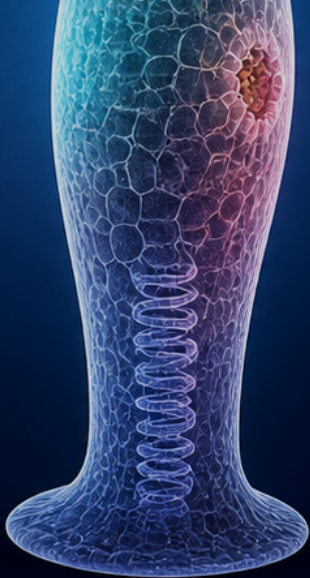
Local feedback interacts with a global mechanical constraint



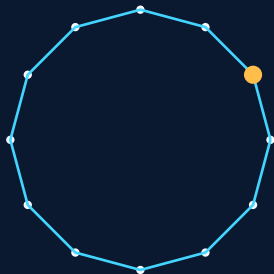
- **Local activation**
Morphogen changes elasticity and concentrates strain locally
- **Long-range inhibition**
The closed pressurized tissue shares a finite total deformation
- **Minimal ingredients**
Morphogen dynamics mechanics and one global constraint

From springs to a continuum

A closed ring of springs reveals how local elasticity produces a global redistribution of strain



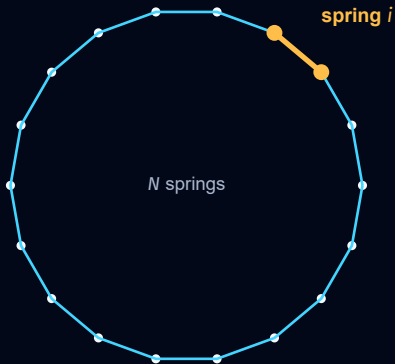
The spring application makes the mechanics tangible



LIVE SPRING MODEL

- **Move one node**
A local displacement changes several neighbouring springs
- **Change one stiffness**
A softer spring takes a larger share of the deformation
- **Close the ring**
The total extension cannot be assigned independently at every point

A ring contains many small springs



$$\Delta x = \frac{L}{N}$$

$$x_i = i\Delta x$$

$$i = 0, \dots, N - 1$$

Every spring represents a segment of material

Compare the reference and deformed spring



extension of spring $i = \ell_i - \Delta x$

Strain measures relative extension

Absolute extension depends on the size of the spring

$$\varepsilon_i = \frac{\ell_i - \Delta x}{\Delta x}$$

- $\varepsilon_i = 0$
The spring keeps its reference length
- $\varepsilon_i > 0$
The spring is stretched
- $\varepsilon_i < 0$
The spring is compressed

The energy of one spring has the correct continuum scaling

Hooke's law assigns the spring energy

$$E_i = \frac{1}{2} k_i(u_i) (\ell_i - \Delta x)^2$$

$$k_i(u_i) = \frac{E(u_i)}{\Delta x}$$

$$E_i = \frac{1}{2} E(u_i) \varepsilon_i^2 \Delta x$$

- **Quadratic response**

Doubling the extension multiplies the elastic energy by four

- **Consistent subdivision**

Increasing the number of springs does not change the tissue stiffness

The factor Δx prepares the energy for a Riemann-sum limit

Morphogen concentration controls local stiffness

$$E'(u) < 0$$

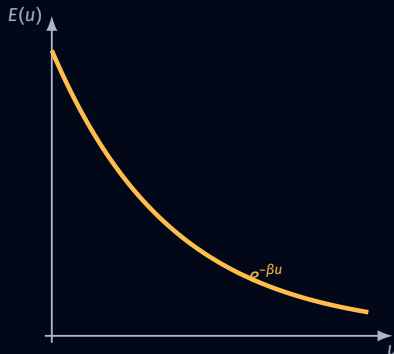
High morphogen

The elastic modulus is smaller and the tissue is softer

• Low morphogen

The elastic modulus is larger and the tissue is stiffer

$$E(u) = e^{-\beta u} \quad \beta > 0$$



The local feedback is positive

higher u $\Rightarrow$ smaller $E(u)$ $\Rightarrow$ larger strain $\Rightarrow$ more u

- **Local activation**

A small local increase in morphogen makes the same region easier to stretch

- **Missing ingredient**

We still need to determine how the strain is distributed around the entire ring

The total discrete energy is a sum over all springs

Every spring contributes its local elastic energy

$$\begin{aligned} E_N &= \sum_{i=0}^{N-1} E_i \\ &= \sum_{i=0}^{N-1} \frac{1}{2} E(u_i) \varepsilon_i^2 \Delta x \end{aligned}$$

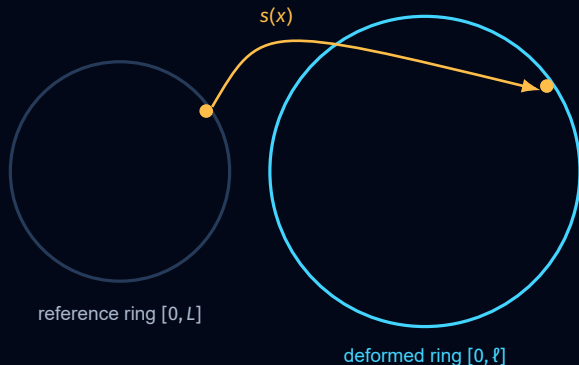
- **Local contribution**

Each term depends only on the strain and morphogen at spring i

- **Global configuration**

The total energy depends on how deformation is shared by all springs

A material coordinate follows the tissue



$$x \in [0, L]$$

$$s(x) \in [0, l]$$

Material label

x identifies a fixed piece of tissue

Deformed position

$s(x)$ gives its arc-length position after stretching

Discrete strain converges to a spatial derivative

$$\ell_i = s(x_{i+1}) - s(x_i)$$
$$\varepsilon_i = \frac{s(x_{i+1}) - s(x_i)}{\Delta x} - 1$$

$$\Delta x \longrightarrow 0 \quad \Rightarrow$$

$$\boxed{\varepsilon(x) = \partial_x s(x) - 1}$$

The strain compares the local deformed length with the reference length

The spring sum becomes a continuum energy

$$E_N = \sum_{i=0}^{N-1} \frac{1}{2} E(u_i) \varepsilon_i^2 \Delta x$$

$$N \longrightarrow \infty \quad \Downarrow$$

$$E_{\text{el}}[s; u] = \frac{1}{2} \int_0^L E(u(x)) \varepsilon(x)^2 dx$$

The discrete spring model has become a continuum elastic energy

The closed ring imposes a global length constraint

$$\begin{aligned}\ell - L &= \sum_{i=0}^{N-1} (\ell_i - \Delta x) \\ &= \sum_{i=0}^{N-1} \varepsilon_i \Delta x\end{aligned}$$



$$\ell - L = \int_0^L \varepsilon(x) dx$$

- Total extension is prescribed**

Inflation determines how much longer the deformed ring is

- Local strains are coupled**

More extension in one region leaves less extension for the rest of the ring

Mechanical equilibrium minimizes elastic energy

Mechanics relaxes rapidly compared with morphogen dynamics

$$s \in \arg \min \left\{ \frac{1}{2} \int_0^L E(u) \varepsilon^2 dx \right\}$$

$$\varepsilon = \partial_x s - 1$$

$$\int_0^L \varepsilon dx = \ell - L$$

- **Quasi-static mechanics**

For each current morphogen profile the tissue immediately reaches mechanical equilibrium

Perturb the deformation to compute the first variation

$$s_\theta(x) = s(x) + \theta h(x)$$

$$\begin{aligned}\varepsilon_\theta &= \partial_x s_\theta - 1 \\ &= \varepsilon + \theta \partial_x h\end{aligned}$$

The periodic perturbation h preserves the closed geometry

$$E_{\text{el}}[s_\theta; u] = \frac{1}{2} \int_0^L E(u) (\varepsilon + \theta \partial_x h)^2 dx$$

$$\begin{aligned}\delta E_{\text{el}} &= \left. \frac{d}{d\theta} E_{\text{el}}[s_\theta; u] \right|_{\theta=0} \\ &= \int_0^L E(u) \varepsilon \partial_x h dx\end{aligned}$$

Integration by parts gives mechanical equilibrium

$$\begin{aligned}\delta E_{\text{el}} &= \int_0^L E(u)\varepsilon \partial_x h \, dx \\ &= - \int_0^L \partial_x (E(u)\varepsilon) h \, dx\end{aligned}$$

The first variation vanishes for every admissible perturbation h

$$\partial_x (E(u)\varepsilon) = 0$$

Mechanical equilibrium determines the strain explicitly

$$E(u(x, t))\varepsilon(x, t) = \sigma(t)$$

$$\varepsilon(x, t) = \frac{\sigma(t)}{E(u(x, t))}$$

$$\ell - L = \sigma(t) \int_0^L \frac{1}{E(u(y, t))} dy$$

$$\varepsilon(x, t) = \frac{\ell - L}{E(u(x, t)) \int_0^L \frac{1}{E(u(y, t))} dy}$$

The full model couples dynamics and mechanical equilibrium

$$\partial_t u = D \partial_{xx} u - \alpha u + \kappa \varepsilon$$

$$0 = \partial_x (E(u) \varepsilon)$$

$$\ell - L = \int_0^L \varepsilon(x, t) dx$$

supplemented with periodic boundary conditions and an initial morphogen profile

- **One evolution equation**
Only the morphogen has its own time dynamics
- **Instantaneous mechanics**
Strain is determined by equilibrium at every time
- **One global constraint**
The prescribed total extension couples the entire ring

Mechanical variables can be eliminated exactly

$$\partial_t u = D \partial_{xx} u - \alpha u + \kappa \varepsilon$$

$$\varepsilon(x, t) = \frac{\ell - L}{E(u(x, t)) \int_0^L \frac{1}{E(u(y, t))} dy}$$



$$\partial_t u = D \partial_{xx} u - \alpha u + \kappa \frac{\ell - L}{E(u(x, t)) \int_0^L \frac{1}{E(u(y, t))} dy}$$

Exponential softening produces the nonlocal feedback

$$E(u) = e^{-\beta u}$$

$$\frac{1}{E(u)} = e^{\beta u}$$

Local factor

$e^{\beta u(x,t)}$ amplifies regions with larger morphogen

$$\varepsilon(x, t) = (\ell - L) \frac{e^{\beta u(x,t)}}{\int_0^L e^{\beta u(y,t)} dy}$$

Global factor

The denominator preserves the prescribed total extension

The mechanics generates a local exponential activation divided by a global normalization

Nondimensionalization gives one nonlocal equation

Natural variables

$$\tilde{x} = \frac{x}{L}$$

$$\tilde{t} = \alpha t$$

$$\tilde{u} = \beta u$$

$$\partial_t u = D \partial_{xx} u - u + \kappa \frac{e^{u(x,t)}}{\int_0^1 e^{u(y,t)} dy}$$

The tildes are dropped and the domain is the unit circle $\mathbb{T} = [0, 1]$

$$u(0, t) = u(1, t)$$

$$\partial_x u(0, t) = \partial_x u(1, t)$$

$$u(x, 0) = u_0(x)$$

One equation realizes the activator–inhibitor paradigm

LOCAL ACTIVATION

$$e^{u(x,t)}$$

A region with more morphogen becomes softer and receives more strain and even production

/

LONG-RANGE INHIBITION

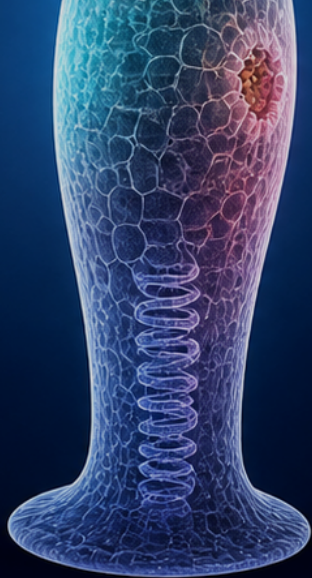
$$\int_0^1 e^{u(y,t)} dy$$

Increased production in one region reduces the share of extension available elsewhere

Mechanical equilibrium converts a local positive feedback into global competition

Steady states and stability

Variational structure determines when patterns exist and which stationary profiles can be stable



Stationary patterns solve a nonlocal boundary-value problem

$$0 = D \partial_{xx} U - U + \kappa \frac{e^{U(x)}}{\int_0^1 e^{U(y)} dy}$$

$$U(0) = U(1)$$

$$\partial_x U(0) = \partial_x U(1)$$

- Homogeneous solution**

$\bar{U} \equiv \kappa$ is always a stationary solution

- Patterned solution**

A nonconstant periodic function $U(x)$ represents a spatial pattern

Every stationary solution has mean value $\int_0^1 U(x) dx = \kappa$

Small diffusion permits patterns while large diffusion suppresses them

THEOREM

For every $\kappa > 0$ there exist constants

$$0 < D_{\min}(\kappa) < D_{\max}(\kappa)$$

- $0 < D < D_{\min}(\kappa)$

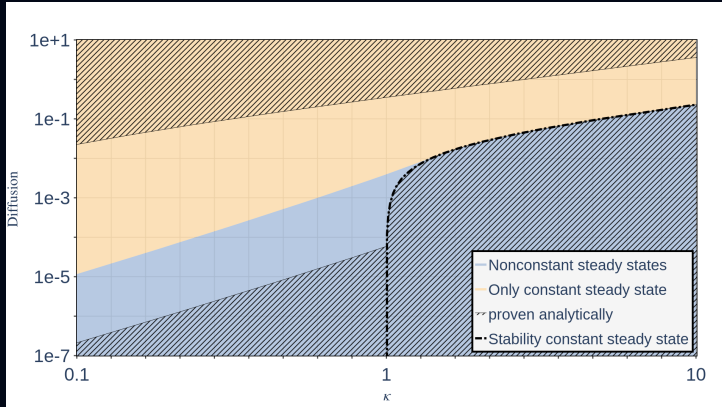
A nonconstant stationary solution exists and is Lyapunov stable in L^2

- $D > D_{\max}(\kappa)$

$\bar{U} \equiv \kappa$ is the unique stationary solution and is globally asymptotically stable

The analytic thresholds are sufficient bounds and need not be optimal

The parameter plane separates homogeneous and patterned regimes



- **Patterns**
Nonconstant steady states are found for sufficiently small diffusion
 - **Homogeneous state**
Large diffusion removes spatial variation
- Hatched regions are established analytically

Linearization asks whether a stationary profile amplifies perturbations

$$u(x, t) = U(x) + e^{\nu t} \varphi(x)$$

Keeping only first-order terms gives the nonlocal eigenvalue problem

$$D\partial_{xx}\varphi + \left(\kappa \frac{e^U}{\int_0^1 e^U} - 1 - \nu \right) \varphi = \kappa \frac{e^U}{\left(\int_0^1 e^U \right)^2} \int_0^1 e^U \varphi$$

- $\nu < 0$

The corresponding perturbation decays

- $\nu > 0$

The corresponding perturbation grows

A local problem helps locate the nonlocal eigenvalues

NONLOCAL PROBLEM

$$D\partial_{xx}\varphi + (A - \nu)\varphi = MC(x) \int_0^1 C(y)\varphi(y) dy$$

Every value of φ is coupled through the integral

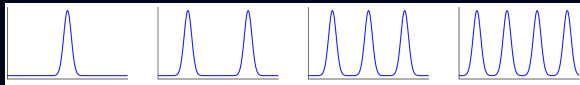
LOCAL PROBLEM

$$D\partial_{xx}\psi + (A - \lambda)\psi = 0$$

Classical oscillation theory orders the eigenfunctions by their

Relevant nonlocal eigenvalues lie between eigenvalues of the associated local problem

Stable mechanochemical patterns are necessarily unimodal



THEOREM

- **Stable nonconstant solution**

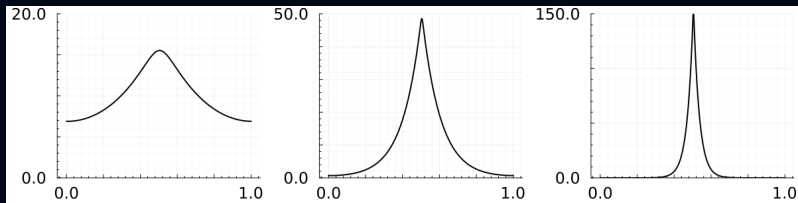
The stable solution obtained by energy minimization has one peak

- **Every multimodal solution**

All solutions with two or more peaks are linearly and nonlinearly unstable

The global mechanical constraint strongly selects a single organizer

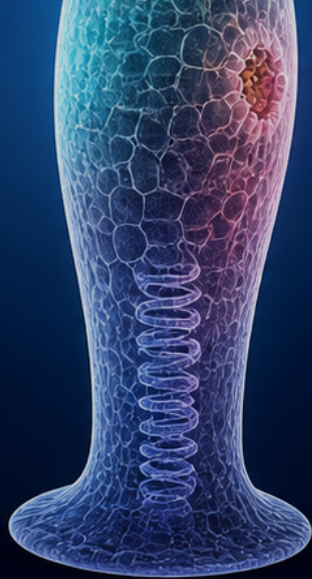
Diffusion changes the width but not the number of stable peaks



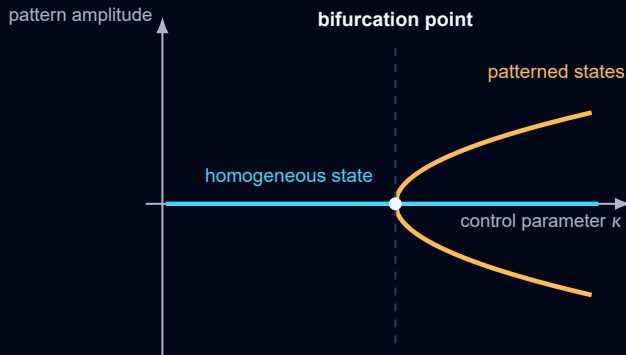
- **Smaller diffusion**
The stable peak becomes sharper and more localized
- **Larger diffusion**
The stable profile becomes broader
- **Robust selection**
The stable pattern retains a single dominant peak

Bifurcations and oscillations

Stationary branches emerge when a homogeneous state changes stability and can generate tissue oscillations



A bifurcation marks a qualitative change in stationary states



- **Before the threshold**
Only the homogeneous stationary state is visible nearby
- **At the threshold**
A spatial mode becomes neutral and a patterned branch can emerge
- **After the threshold**
The number or stability of stationary states has changed

The homogeneous state loses stability at discrete thresholds

$$\bar{U} \equiv \kappa$$

$$\varphi_n(x) = \cos(2\pi nx)$$

Linearization assigns one growth rate to every spatial mode

$$v_n = \kappa - 1 - 4\pi^2 n^2 D$$

$$\kappa_n = 1 + 4\pi^2 n^2 D$$

- $n = 1$

The first nonconstant mode has one maximum and one minimum

- $n = 2$

The next mode produces a two-peak spatial structure

- **Discrete thresholds**

Each mode changes stability at its own value of κ

The first patterned branch can be subcritical or supercritical

THEOREM

At $\kappa = \kappa_n$ a branch of nonconstant stationary solutions bifurcates from the homogeneous state $\bar{U} \equiv \kappa$

- **Subcritical branch**

$\kappa_n < 3/2$ equivalently $D < (8\pi^2 n^2)^{-1}$

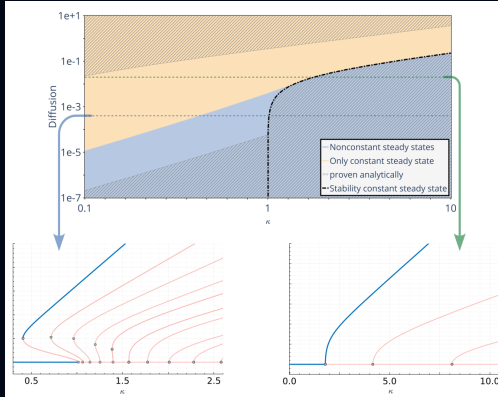
- **Supercritical branch**

$\kappa_n > 3/2$ equivalently $D > (8\pi^2 n^2)^{-1}$

Near the bifurcation only the supercritical branch generated by the first mode can be stable

The direction of the branch determines whether small patterned states appear before or after the linear threshold

Diffusion determines the type of each bifurcation



- **Small diffusion**
The branch can bend toward smaller values of κ
- **Larger diffusion**
The branch can emerge toward larger values of κ

Each spatial mode has its own transition curve

Tissue length turns the stationary picture into dynamics

Let $\ell(t)$ denote the current circumference and L its reference value

$$\partial_t u = D \partial_{xx} u - u + \kappa (\ell(t) - L) \frac{e^{u(x,t)}}{\int_0^1 e^{u(y,t)} dy}$$

- Morphogen rises**

The tissue softens and the mechanical equilibrium shifts

- The spheroid expands**

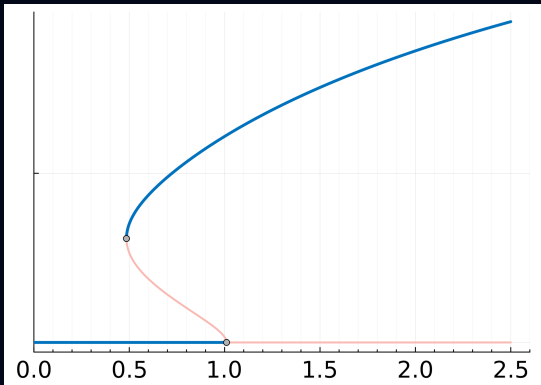
Increased strain strengthens morphogen production

- Feedback relaxes**

Degradation and mechanical restoration can reverse the expansion

Coupling pattern formation to tissue size creates a route from static symmetry breaking to oscillations

Subcritical feedback permits abrupt switching



- **Coexisting states**

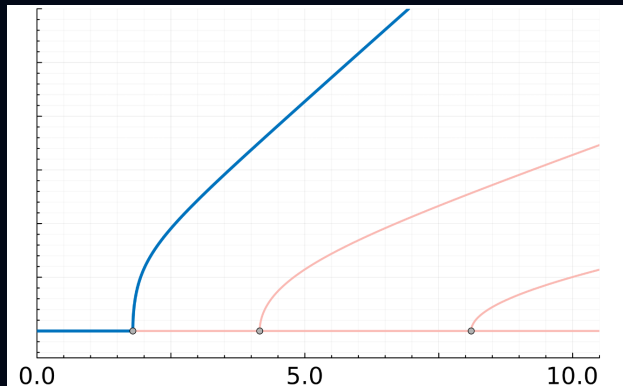
A homogeneous and a patterned state may be available for the same parameter value

- **Sudden transition**

Crossing a fold can force the system to jump to a distant branch

simulation placeholder
abrupt inflation and relaxation

Supercritical feedback produces a gradual patterned response



- Continuous onset**

The patterned amplitude grows continuously from the homogeneous state

- Smooth mechanical response**

Tissue deformation can track the changing stationary branch

simulation placeholder
gradual inflation and pattern growth

Mechanics can realize an activator–inhibitor mechanism without a chemical inhibitor

- **Local positive feedback**

Stretch increases morphogen production and morphogen softens the tissue

- **Global mechanical competition**

The fixed total length couples every point through one nonlocal constraint

- **Single organizer**

Stable nonconstant stationary states contain one dominant peak

morphogen $\Rightarrow$ softening $\Rightarrow$ strain $\Rightarrow$ morphogen

A single morphogen equation coupled to mechanics can implement local activation and long-range inhibition